TagParaFormer: Cross-Hybrid Attention Learning Framework for Topology-Aware Road Network Extraction From Remote Sensing Imagery

Boaz Mwubahimana, *Graduate Student Member, IEEE*, Dingruibo Miao, Yan Jianguo, Le Ma, *Member, IEEE*, Remy Dukundane, Xiao Huang, Swalpa Kumar Roy, *Senior Member, IEEE*, and Ruisheng Wang, *Senior Member, IEEE*

***Abstract*—Autonomous high-resolution remote sensing (RS) imagery provides fine-grained spatial and semantic cues essential for reliable road network extraction, a task fundamental to transportation planning, navigation systems, and infrastructure management. Despite advances in convolutional neural networks (CNNs) and transformers, existing approaches frequently fail to preserve topological continuity at intersections and occluded segments, where structural gaps and disconnections arise even when pixelwise accuracy is high. This limitation stems from the inherent difficulty of modeling graph-structured connectivity using pixelwise supervision alone. To address these challenges, we reformulate road extraction as a topology-preserving representation learning problem and introduce TagParaFormer, a topology-aware parallel dual-stream CNN–vision transformer (ViT) framework with an explicit graph-based topology refinement strategy. The proposed architecture integrates an attention node graphical (ANG) module for explicit node-level junction modeling, a road refinement backbone (RRB) for multiscale edge-continuity enforcement, and a Bayesian Parzen-tree estimator (PTE) for data-adaptive topology-sensitive hyperparameter optimization. Network fragmentation, defined as the ratio of disconnected road segments to total predicted road, is substantially reduced against both conventional and dedicated topology-aware baselines, confirming that improvements are rigorous and general. Extensive experiments on four benchmark datasets demonstrate consistent state-of-the-art performance in connectivity rate (CR), gap-free rate (GR), and junction accuracy, achieving competitive efficiency with substantially fewer parameters than transformer-heavy architectures. The codebase is at GitHub.**

***Index Terms*—Bayesian optimization, convolutional neural networks (CNNs), graph attention network, remote sensing (RS), road network extraction, topology-preserving segmentation, vision transformer (ViT).**

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Boaz Mwubahimana, Dingruibo Miao, and Le Ma are with the State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing (LIESMARS), Wuhan University, Wuhan 430079, China (e-mail: m.boaz@whu.edu.cn; miaodrb@whu.edu.cn; malele@whu.edu.cn).

Yan Jianguo is with LIESMARS, Wuhan University, Wuhan 430079, China, and also with Xinjiang Astronomical Observatory, Chinese Academy of Sciences, Ürümqi 830011, China (e-mail: jgyan@whu.edu.cn).

Remy Dukundane is with the College of Engineering, Carnegie Mellon University Africa, Kigali, Rwanda (e-mail: rdukunda@andrew.cmu.edu).

Xiao Huang is with the Department of Environmental Sciences, Emory University, Atlanta, GA 30322 USA (e-mail: xiao.huang2@emory.edu).

Swalpa Kumar Roy is with the Department of Computer Science and Engineering, Tezpur University, Tezpur, Assam 784028, India (e-mail: swalpa@tezu.ernet.in).

Ruisheng Wang is with the School of Architecture and Urban Planning, Shenzhen University, Shenzhen 518060, China (e-mail: ruiswang@szu.edu.cn).

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I. Introduction

ACURATE extraction of road networks from very high-resolution (VHR) remote sensing (RS) imagery is fundamental to numerous real-world applications [1], [2], [3], including autonomous navigation, transportation management, urban planning, and routine updates of geographic information system (GIS) databases [4], [5], [6]. Although recent advances in deep-learning-based semantic segmentation [7] have been able to improve the visual fidelity of road predictions, the extracted road networks still frequently exhibit topological inconsistencies, including gaps, disconnections, and spurious links, which substantially degrade their usability in downstream applications. As Fig. 1(d) reports, such errors persist even when pixelwise segmentation accuracy appears high. This limitation primarily stems from the formulation of road extraction as a per-pixel classification problem, where connectivity constraints and the inherently graph-structured topology of road networks are not explicitly modeled. For instance, Hu et al. [8] employed an enhanced watershed algorithm, while He et al. [9] introduced template-matching strategies for linear feature detection. Lijun et al. [10] leveraged Canny edge detection for automated extraction. Despite their intuitive design, such handcrafted methods are highly sensitive to local variations and prone to failure in complex urban scenes where occlusions from vehicles, shadows, and built structures degrade visibility [11], [12].

The use of deep convolutional neural networks (CNNs) has recently transformed road extraction. Architectures such as U-Net [13] and its successors (e.g., LinkNet [14], D-LinkNet [15], and DenseUNet [16]) employ encoder–decoder structures with skip connections to preserve fine spatial detail. Attention-based designs further enhance feature discrimination by emphasizing informative regions [17].

Deep learning approaches for road extraction have evolved substantially, with increasing network depth generally yielding improved performance [18], [19]. However, training very deep architectures presents challenges such as vanishing gradi-

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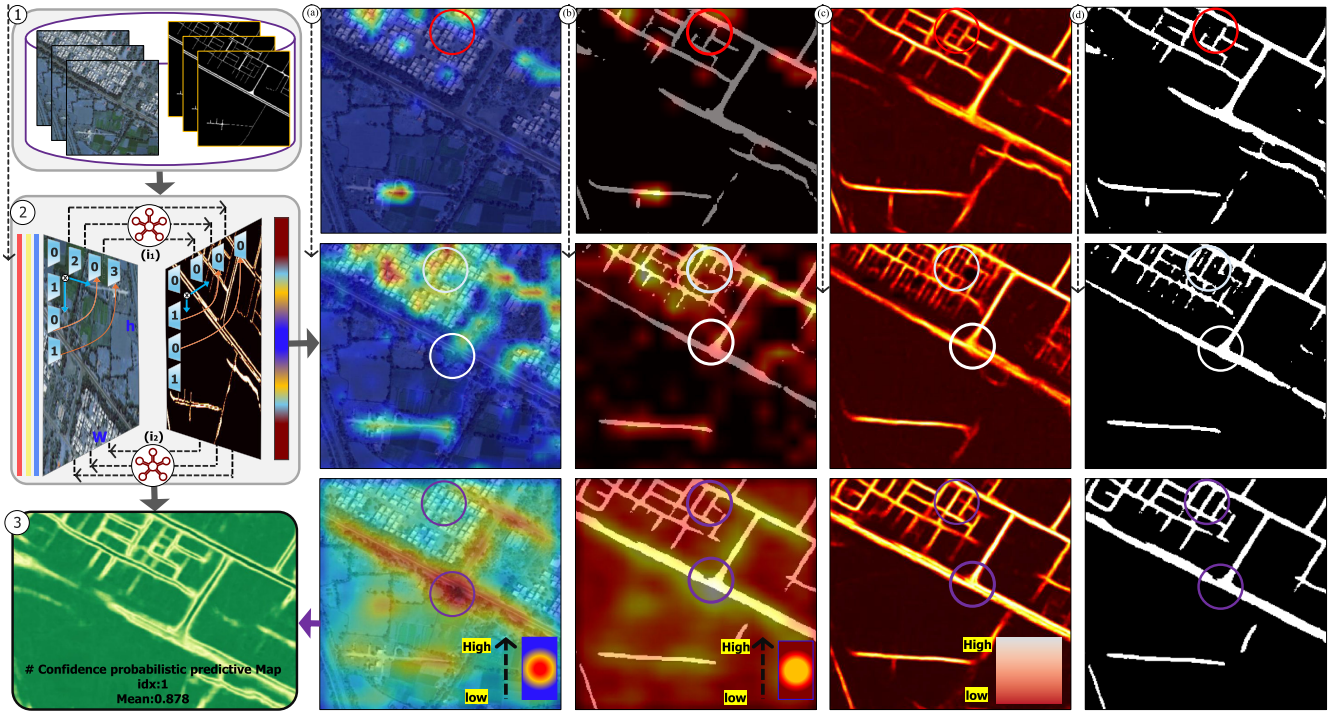


Fig. 1. Graphical abstract. 1—input data. 2—semantic graph node relations. (a) Grad-CAM geometry nodes. (b) Multilevel semantic alignment. (c) Nodal probability maps. (d) Prediction threshold maps. 3—confidence maps: red = disconnected; white = partially improved; purple = reconnected after refinement i_1, i_2.

ents. To mitigate this issue, He et al. [20] introduced deep residual learning with identity mapping [21], enabling successful training of substantially deeper networks. In contrast to simple skip connections in fully convolutional networks, Ronneberger et al. [13] presented the U-Net, which concatenates feature maps from different levels to enhance segmentation accuracy by integrating low-level detail information with high-level semantic information. This architecture has demonstrated particularly strong results in biomedical image segmentation and has been widely adapted for road extraction tasks [14].

Given the advantages of deep learning in efficiency, effectiveness, and broad adaptability, numerous studies have focused on applying these techniques to road extraction. To address network depth-related challenges, He et al. [20] proposed residual networks (ResNet) to counteract potential network degradation and gradient vanishing during training. Subsequently, LinkNet [14] combined aspects of both U-Net and ResNet, modifying the feature combination strategy from concatenation to skip connections. Huang et al. [22] introduced DenseUNet, incorporating dense connections and skip connections to handle occlusions from trees and shadows [23], [24], [25]. Despite these contributions, a common drawback in such architectures is the application of successive downsampling in initial layers, converting input information into fixed-size representations, thereby causing substantial information loss. Although CNNs have achieved remarkable success in segmentation tasks, they are limited by their localized receptive fields, which prevent effective modeling of global contextual relationships, a critical requirement for road network topology [26]. Consequently, Dosovitskiy et al. [6] proposed vision transformer (ViT) in 2020 for global image recognition, leveraging self-attention mechanisms to capture long-range dependencies. In our TagParaFormer framework, the ViT branch specifically addresses the challenge of modeling global node connectivity patterns essential for maintaining topological consistency. Through parallel processing with the CNN branch, each pathway undergoes feature refinement: local features are enriched with global context [27], while semantic feature maps are optimized via Bayesian optimization to detect isolated nodes. The Parzen-tree estimator (PTE) further refines predictions to achieve optimal topological performance [22].

Taking advantage of global self-attention, ViTs [18] have demonstrated strong capability in modeling long-range dependencies, inspiring hybrid CNN–ViT systems [28], [29], [30], [31], [32], [33] that unify local geometric cues with global contextual reasoning.

Despite these architectural advances, however, existing methods predominantly focus on improving pixel-level metrics (e.g., IoU and $F1$-score) without explicitly considering the structural and relational constraints inherent to road networks. As illustrated in Fig. 1(a) and (d) of the graphical abstract and Fig. 2 of the CNN branch 2, road topology is inherently graph-structured: nodes represent intersections or endpoints, while edges correspond to road segments. Topological errors typically manifest as follows.

1) *Gaps:* Physically continuous roads predicted as disconnected segments.
2) *Misconnections:* Incorrect links between unrelated roads.
3) *Isolated Nodes:* Missing or improperly modeled intersections.

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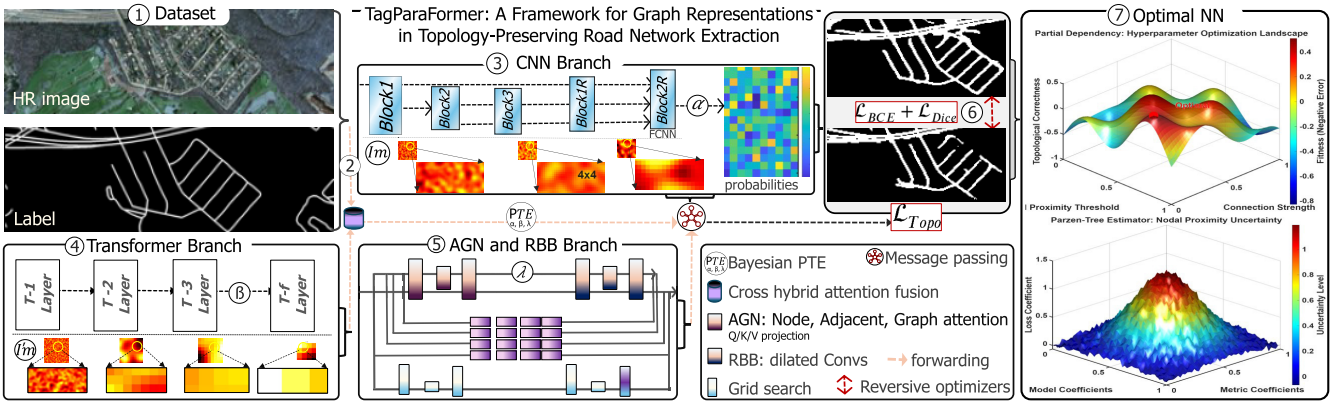


Fig. 2. Overview of TagParaFormer.

TABLE I

ARCHITECTURAL NOVELTY COMPARISON. ✓ = SUPPORTED; P = PARTIAL; × = NOT SUPPORTED. E2E: END-TO-END; ND: NODE DETECTION; DA: DYNAMIC ADJACENCY; PS: PARALLEL CNN–VIT; BO: BAYESIAN OPTIMIZATION; TL: TOPOLOGY-AWARE LOSS

Method	E2E	ND	DA	PS	BO	TL
Sat2Graph	×	P	×	×	×	×
RNGDet++	P	P	×	×	×	×
BT-RoadNet	×	×	×	×	×	×
SAM-Road++	✓	×	×	×	×	×
MagRoad	✓	×	×	×	×	P
Swin-UNet	✓	×	×	×	×	×
Ours	✓	✓	✓	✓	✓	✓

TABLE II

TOPOLOGICAL METRICS COMPARISON ACROSS SOA METHODS ON DEEPGLOBE DATASET

Method	Conn.	Gf	J	C	Node
RNGDet++	72.3	68.4	70.2	71.8	69.5
Sat2graph	74.1	70.2	72.4	73.5	71.2
AttUNet	78.6	74.2	76.1	77.3	75.8
MagRoad	76.8	72.5	74.3	75.6	73.4
D-LinkNet	81.4	79.3	80.5	82.1	79.6
MaGRoad	80.2	77.8	79.1	80.4	78.3
ViT-B/16	82.5	80.1	81.3	82.9	80.7
Swin Transformer	83.7	81.4	82.6	84.1	81.9
Swin-UNet	86.2	84.5	85.3	87.2	84.3
BT-RoadNet	84.3	82.1	83.4	84.8	82.5
SAM-Road	85.7	83.8	84.2	86.5	83.1
Ours	**94.5**	**92.8**	**91.3**	**93.7**	**89.8**

Notes: Conn. = Connectivity rate; Gf = Gap-free rate; J = Junction Accuracy; C = Centerline Precision; Node = Node Recall.

It should be noted that such errors cannot be resolved through pixel-level supervision alone, as they require explicit modeling of connectivity relationships and structural coherence, a capability ignored by conventional segmentation objectives.

To this end, we propose TagParaFormer, a dual-branch framework that jointly models local road geometry and global topology through CNN–ViT fusion and explicit topological reasoning modules. TagParaFormer addresses road fragmentation by learning to detect and correct connectivity errors directly within the segmentation process.

The key technical contributions of this work can be summarized as follows.

1) We propose a parallel dual-stream CNN–ViT topology-preserving representation learning framework, termed TagParaFormer, with graph-guided edge refinement for end-to-end road network extraction.
2) We introduce the attention node graphical (ANG) module and road refinement backbone (RRB). ANG detects road nodes via a learnable two-layer MLP, constructs a dynamic adjacency matrix, and propagates corrections through graph attention. RRB applies parallel dilated convolutions for multiscale edge-continuity refinement, distinguishing it from uniform segmentation heads and postprocessing corrections.
3) We incorporate a Bayesian PTE that adaptively tunes λ, α, and β (to be defined in Section III) in a data-driven manner, with parameter sensitivity analysis for a flat, robust optimal region (see Table VIII).
4) We provide empirical validation, including controlled cross-removal ablation and architectural comparison against all topology-aware baselines, through experiments conducted on diverse benchmarks.

A comprehensive architectural comparison against all topology-aware baselines is summarized in Table I. The remainder of this article is structured as follows. Section III details the TagParaFormer architecture and modules. Section IV describes the datasets, implementation settings, and evaluation metrics. Section IV-A presents experimental results, Section V includes experimental, computational complexity, limitations, topological, and ablation studies analysis. Finally, Section VI concludes the study and discusses limitations and future perspectives.

II. RELATED WORKS

A. CNN-Based and Hybrid Segmentation

Beyond the encoder–decoder designs noted in Section I, SegNet [34] and DeepLabv3 + [33] contributed atrous convolutions for multiscale context. In RS, FWDNNet [27] introduced tensor-based fusion for heterogeneous encoders [35], [36], and Guo et al. [4] explored building-road collaborative extraction. Hybrid CNN–ViT designs [18], [28], [30], [31], [32] further combine local geometry with global context, yet one thing in common is that all predominantly optimize pixelwise metrics (IoU and $F1$) without explicitly

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TABLE III
DATASET CHARACTERISTICS AND DOMAIN VARIATIONS

Dataset	Resolution	Coverage Area	Spatial Coverage	Urban/Rural	Season	Image Count	Annotation Type
DeepGlobe	0.5 m	2,220 km²	Thailand, Indonesia, India	Mixed	Dry	6,226	Binary road masks
Massachusetts	1.2 m	2,600 km²	Massachusetts, USA	Suburban	Multiple	1,108	Binary segmentation
SpaceNet	0.31 m	3,450 km²	Las Vegas, Paris, Shanghai, Khartoum	Urban	Varies	2,780	Road centerlines (GeoJSON)
WHU	0.075 m	20,000 km²	Christchurch, New Zealand	Urban	Summer	6,471	Binary segmentation

TABLE IV
COMPARATIVE TRAINING BEHAVIOR ACROSS ALL MODELS AND DATASETS WITH ALL METRICS

No	Architecture	DeepGlobe									Massachusetts									SpaceNet								
		L	A	P	R	U	mU	F1	B1	C	L	A	P	R	U	mU	F1	B1	C	L	A	P	R	U	mU	F1	B1	C
1	**TagParaFormer**	**16**	**98**	**91**	**89**	**84**	**92**	**90**	**89**	**85**	32	97	82	80	72	88	81	79	76	**29**	**96**	86	83	**76**	89	**84**	82	79
2	SAM-Road	27	88	84	82	84	88	90	82	79	35	87	83	81	75	85	83	80	77	31	87	83	81	75	86	84	81	78
3	Swin-UNet	31	87	83	81	83	87	90	81	78	36	86	82	80	74	84	83	79	76	32	85	82	80	75	85	84	80	77
4	MagRoad	39	72	72	70	39	72	56	53	51	41	71	71	69	40	71	55	52	50	40	70	70	68	39	70	54	51	49
5	Sat2Graph	32	87	83	81	81	85	89	82	79	33	85	82	80	72	83	82	81	78	30	84	83	80	74	84	83	80	77
6	ViT-B/16	76	51	62	60	60	72	61	59	56	75	50	61	59	59	71	61	58	55	74	48	60	58	58	70	60	57	54
7	SAM-Road++	39	88	79	77	69	80	79	75	71	37	86	78	76	68	79	77	73	70	36	85	79	77	70	80	78	75	71
8	RNGDet++	15	82	73	71	63	75	72	69	65	32	80	72	70	61	74	70	68	64	31	79	73	71	62	75	71	69	65
9	AttUNet	54	77	75	73	70	78	82	73	70	53	74	74	72	67	77	80	72	69	52	73	75	73	66	78	79	73	70
10	Sat2graph	64	83	82	80	76	82	86	79	76	62	81	81	79	74	82	85	78	75	61	80	81	79	73	83	84	79	76

Notes: **Blue bold** indicates best value. Metrics (×100): L=Loss, A=Avg Acc, P=Precision, R=Recall, U=IoU, mU=mean IoU, F1=F1-Score, B1=BF1, C=Compactness. All values ×100 (e.g., 0.16→16).

modeling road network structure, thereby leaving topological errors unresolved [37].

B. Graph Neural Networks and Topology-Preserving Road Extraction

Recognizing that road networks are graph-structured, with nodes as junctions and edges as segments, recent work shifted toward explicit topology modeling. Zhou et al. [38] employed GCNs for node-level neighborhood aggregation; Guth et al. [39] provided a taxonomy of topological errors (gaps, misconnections, and isolated nodes) used as evaluation criteria in this work; and dual-task networks [32], [40] jointly optimize segmentation and centerline detection, yet lack node-level connectivity modeling.

Among dedicated topology-preserving methods: RNGDet++ [41] uses reinforcement-learning-based iterative graph detection but requires a two-stage pipeline; BT-RoadNet applies a bifurcation-tree structure as postsegmentation correction rather than an integral training objective; Sat2Graph [42] infers graphs via recurrent traversal on pre-extracted skeletons, yielding strong metrics at high computational cost; SAM-Road/SAM-Road++ [43], [44] encode topology implicitly via foundation-model priors; and MagRoad [45] adds connectivity cues without junction-level graph structure. All of these either require postprocessing, operate on pre-extracted representations, or treat topology as secondary to pixel accuracy. TagParaFormer instead integrates node detection, dynamic adjacency construction, and graph-attention propagation within a unified end-to-end encoder–decoder with topology-aware losses, a design absent in all prior works (see Table II).

III. METHODOLOGY

A. Network Overview

We have developed TagParaFormer as a topology-aware encoder–decoder (presented in Figs. 1 and 2 with $i_{①}$, $i_{②}$ graph message passing iterations) that integrates heterogeneous CNN and Transformer branches through graph-based refinement, as illustrated in Fig. 2. Specifically, we have designed four core components: 1) a dual-branch parallel encoder for joint local–global feature extraction shown in Figs. 3 and 4; 2) an ANG module for explicit node-level topology modeling; 3) an RRB for edge-level structural correction; and 4) a Bayesian PTE for connectivity-sensitive hyperparameter optimization.

Conventionally, architectures combine CNNs and transformers through simple concatenation or feature summation. On the other hand, these approaches fail to preserve topological continuity at intersections and occluded segments where roads fragment. Even though some works attempt postprocessing refinement, they remain limited in modeling the inherent graph structure of road networks, nodes as junctions, and edges as road segments. To address these limitations, we have integrated graph-aware modules that explicitly model node proximity and adjacency relations.

We further combine two branches as Fig. 2 depicts. The CNN branch extracts local geometric features $F_{cnn} = E_{cnn}(\mathcal{X})$ through eight residual blocks, while the ViT branch captures long-range dependencies via multihead self-attention to produce $F_{vit} = E_{vit}(\mathcal{X})$. As a result of this parallel processing, geometric precision and topological completeness improve simultaneously.

We also introduce the ANG module to explicitly model road topology through graph-based reasoning. Unlike conventional segmentation heads operating on dense pixel grids, we have designed ANG to identify critical node points—junctions and endpoints—and model pairwise connectivity via attention scores $A_{ij} = \text{softmax}(Q_i K_j^T / \sqrt{d_k})$. Unlike standard attention, which captures pixelwise relationships uniformly, node-level graph reasoning better preserves structural continuity beyond local neighborhoods.

As Fig. 3 shows, the RRB reinforces internode dependencies and corrects structural discontinuities. Even though

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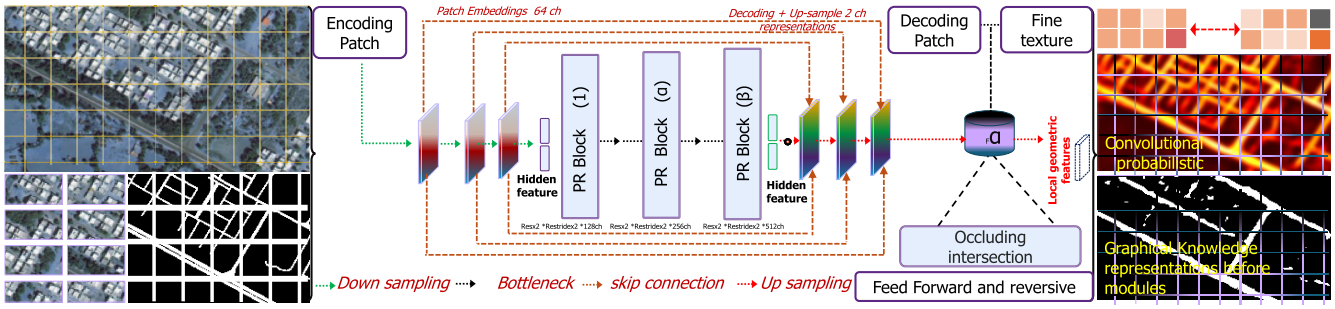


Fig. 3. CNN branch.

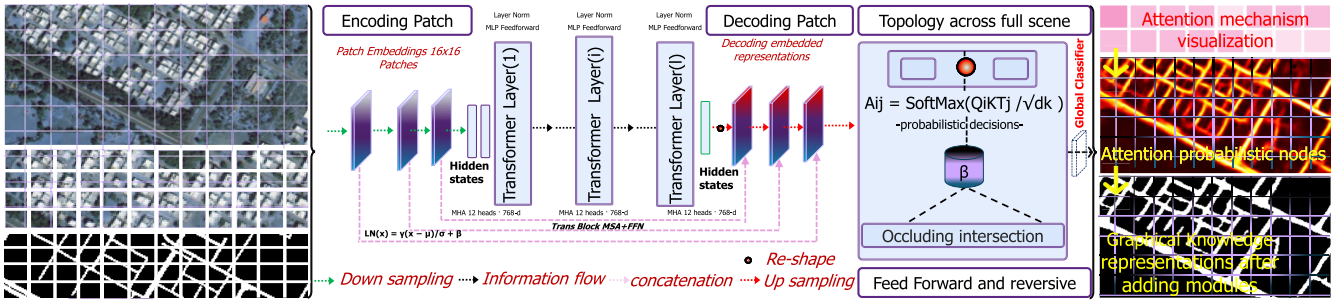


Fig. 4. ViT branch.

naive fusion treats all features equally, edge-level refinement improves geometric regularity. The RRB propagates node-level constraints across the network, reconnecting fragmented segments in hidden features while suppressing fine texture into spurious connections.

The Bayesian PTE optimizes three topology-sensitive hyperparameters: connectivity threshold λ, CNN weight α, and ViT weight β. Through this optimization, balancing local–global contributions minimizes topological degradation while maintaining efficiency.

1) Implementation Details: TagParaFormer is trained using the AdamW optimizer with an initial learning rate of 1×10^{-4}, weight decay 5×10^{-4}, and a cosine annealing schedule over 150 epochs. The batch size is set to 8 on a single NVIDIA A100 GPU. Data augmentation includes random horizontal/vertical flips, random rotation ($\pm 30°$), color jitter (brightness ± 0.2 and contrast ± 0.2), and random cropping to 512×512 patches. The CNN backbone uses ImageNet-pretrained ResNet-50 weights; the ViT encoder is initialized from a ViT-B/16 checkpoint pretrained on ImageNet-21K. Bayesian PTE runs for 50 optimization iterations per dataset.

Fig. 3 displays how TagParaFormer uses a modified 1-D ResNet for node edge detection, replacing 2-D layers with 1-D convolutions for sequential node modeling and redesigning downsampling to improve junction accuracy and gap-free rate (GR) per PTE analysis. The network comprises eight residual blocks (kernels 1, 2, and 4) with batch normalization and ReLU, followed by max pooling. The final block feeds into binary cross-entropy (BCE) loss (L_{bce}) to produce F_b. As illustrated in Fig. 2, TagParaFormer processes inputs through: (top panel) a CNN branch preserving geometric resolution; (down left) a Transformer branch modeling global dependencies; and 5 ANG and RRB modules for node-level graph modeling and edge-level correction. Both ANG and RRB operate as supervised dual-path assisted learning (SDPAL) components, enhanced through improved graph attention network residual learning and double residual convolutions for feature attention.

B. CNN Branch

As can be seen from Fig. 3, the CNN branch uses a modified ResNet with eight residual blocks to extract dense local features from high-resolution satellite imagery. This branch explicitly targets road edges and fine-grained textures, preserving geometric fidelity essential for centerline accuracy but insufficient for long-range topology.

C. Transformer Branch

Furthermore, to complement the CNN's local focus, TagParaFormer's Transformer branch, as shown in Fig. 4, implements a three-layer ViT that captures nonlocal dependencies across the scene. This enables consistent topology at occluded intersections and disconnected segments by globally reasoning about road connectivity, effectively bridging gaps where local features alone fail.

D. Graph Attention Residual Learning (GARL)

Let X be the input feature map, $F_{i,c}(X)$ be the residual function, and $M_{i,c}(X)$ be the attention mask. The output of an attention residual block is

$$H_{i,c}(X) = F_{i,c}(X) \odot M_{i,c}(X) + X$$

where $\odot$ denotes the elementwise multiplication [20], [21], [45]. Alternatively, standard residual learning can be expressed as follows:

$$H_{i,c}(X) = X + F_{i,c}(X). \quad (1)$$

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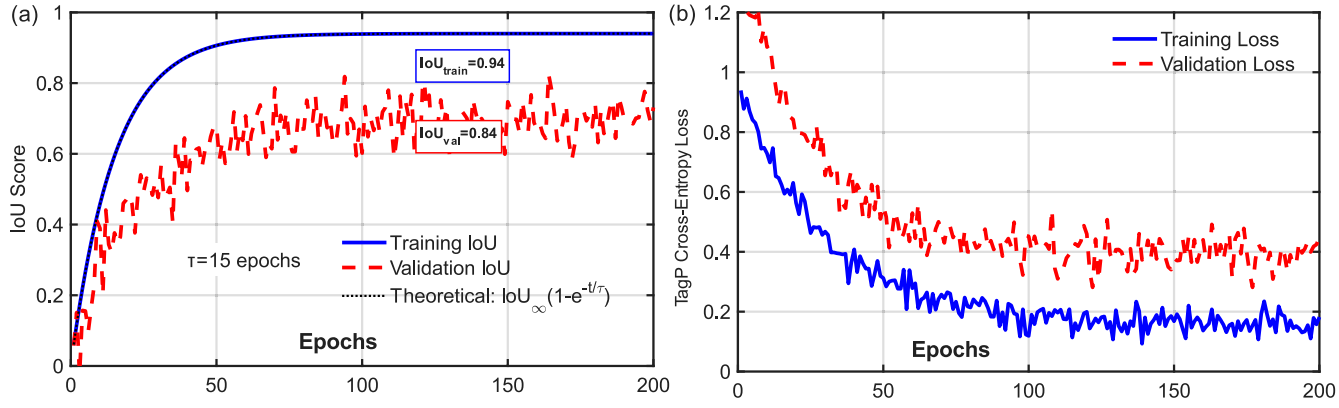


Fig. 5. TagParaFormer historical training behavior over metrics of (a) IoU and (b) TagLoss against epochs.

Algorithm 1 ANG Module

Input: Fused feature map $F_{hybrid} \in \mathbb{R}^{H\times W\times C}$
Output: Topology-corrected feature map F_{topo}

Step 1: Node Detection
$s_i \leftarrow \sigma(\mathrm{MLP}_2(\mathrm{ReLU}(\mathrm{MLP}_1(F^i_{hybrid}))))$; // node confidence
$\mathcal{V} \leftarrow \{i : s_i > 0.5\}$; // detected node set
$h_i \leftarrow W_e \cdot F^i_{hybrid}, \forall i \in \mathcal{V}$; // node embeddings

Step 2: Adjacency Construction
foreach *pair* $(i, j) \in \mathcal{V} \times \mathcal{V}$ **do**
  $A_{ij} \leftarrow \sigma(\mathrm{MLP}([h_i \| h_j \| \|p_i - p_j\|]))$
end foreach

Step 3: Graph Attention Propagation (2 rounds)
for $r = 1, 2$ **do**
  foreach $i \in \mathcal{V}$ **do**
    $\alpha_{ij} \leftarrow \mathrm{softmax}_j\left(\frac{(W_Q h_i)(W_K h_j)^T}{\sqrt{d_k}} \cdot A_{ij}\right)$
    $h_i^{(r)} \leftarrow \mathrm{ReLU}\left(\sum_j \alpha_{ij} W_V h_j^{(r-1)}\right)$
  end foreach
end for

Step 4: Re-projection
$F_{topo} \leftarrow F_{hybrid} + W_{out} \cdot \mathrm{Scatter}(\{h_i^{(2)}\}_{i\in\mathcal{V}})$

Activation functions include

$$\mathrm{ReLU}(k) = \begin{cases} 0, & k \le 0 \\ k, & k > 0 \end{cases}, \quad \mathrm{LReLU}(k) = \begin{cases} k, & k > 0 \\ \alpha k, & k \le 0. \end{cases} \tag{2}$$

Spatial attention uses horizontal and vertical global pooling

$$Z_c^h(h) = \frac{1}{W}\sum_i X_c(h, i), \quad Z_c^w(w) = \frac{1}{H}\sum_j X_c(j, w) \tag{3}$$

concatenated and processed to produce attention weights g^h and g^w, with final graphical output

$$\mathrm{Out} = g^h(i) \times g^w(j). \tag{4}$$

Stacking these attention modules leads to progressive refinement of feature maps, emphasizing informative responses while suppressing erroneous connections.

E. ANG Module

The ANG module addresses the failure of pixelwise methods to model road junction structure. Given the fused feature map F_{hybrid}, a lightweight two-layer MLP classifier produces a node confidence score s_i for each spatial position i. Positions with $s_i > 0.5$ are designated as road nodes: Since the threshold operates on normalized sigmoid scores from learned embeddings rather than raw pixel intensities, it remains resolution-invariant across datasets; scale-specific node density variations are further handled by the Bayesian PTE, which adapts λ per dataset through data-driven optimization and representing intersections or endpoints. Each detected node $i \in \mathcal{V}$ is assigned an embedding $h_i = W_e \cdot F^i_{hybrid}$.

A dynamic adjacency matrix $\mathbf{A}$ is then constructed by jointly encoding embedding similarity and spatial proximity

$$A_{ij} = \sigma\left(\mathrm{MLP}\left(\left[h_i \| h_j \| \|p_i - p_j\|\right]\right)\right) \tag{5}$$

where p_i and p_j are the spatial coordinates of nodes i and j. Two rounds of masked graph attention propagation update each node embedding

$$h_i^{(r)} = \mathrm{ReLU}\left(\sum_j \alpha_{ij} W_V h_j^{(r-1)}\right) \tag{6}$$

$$\alpha_{ij} = \mathrm{softmax}_j\left(\frac{\left(W_Q h_i\right)\left(W_K h_j\right)^T}{\sqrt{d_k}} \cdot A_{ij}\right). \tag{7}$$

Updated embeddings are scattered back to the pixel grid via a linear projection, yielding the topology-corrected map $F_{topo} = F_{hybrid} + W_{out} \cdot \mathrm{Scatter}(\{h_i^{(2)}\})$. Unlike standard attention, which operates uniformly over all pixel pairs, ANG restricts message passing to detected junction nodes and weights edges by structural proximity, ensuring that corrections propagate along physically plausible road paths rather than across irrelevant image regions.

F. Road Refinement Backbone

The RRB enforces structural edge continuity on F_{topo} and explicitly differs from both standard segmentation heads and postprocessing corrections. Standard segmentation heads apply a single uniform convolutional projection to the feature map,

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treating all spatial positions identically. Postprocessing methods operate on binarised output masks outside the training loop. RRB, by contrast, operates on learned feature representations inside the end-to-end training objective and applies structurally differentiated processing at three spatial scales.

Specifically, RRB applies three parallel dilated convolutional branches to F_{topo}

$$F_{d_r} = \mathrm{Conv}_{3\times3}^{\mathrm{rate}=r}\left(F_{topo}\right), \quad r \in \{1,2,4\} \tag{8}$$

where dilation rate $r = 1$ captures fine-grain road edges, $r = 2$ models mid-scale road segments, and $r = 4$ spans wide intersections and where dilation rate $r = 1$ captures fine-grain road edges, $r = 2$ models mid-scale road segments, and $r = 4$ spans wide intersections and coarse structural gaps, a geometric progression empirically confirmed to outperform arithmetic alternatives such as $\{1, 3, 6\}$ that oversample intermediate scales at the cost of edge detail. The three outputs are concatenated and projected

$$F_{refined} = \mathrm{Conv}_{1\times1}\left(\left[F_{d_1} \| F_{d_2} \| F_{d_4}\right]\right). \tag{9}$$

This multiscale dilation design captures road continuity at multiple spatial frequencies simultaneously, guided by the node-level structural priors established by ANG, a design property absent in both uniform segmentation heads and postprocessing refinement pipelines.

G. TagLoss and Optimization

To jointly enforce semantic correctness and structural continuity, the learning objective in TagParaFormer is formulated from a topology-aware representation learning perspective rather than treating predictions as independent pixel events. Although pixelwise supervision is effective for local classification, it is nevertheless insufficient for preserving road connectivity at intersections and occluded segments. The proposed loss integrates complementary objectives defined on the same prediction space to jointly regularize geometry and structure.

A BCE loss is first employed to supervise pixel-level road classification

$$L_{\mathrm{BCE}} = -\sum_{i}\left[y_i \log\left(\hat{y}_i\right) + \left(1 - y_i\right)\log\left(1 - \hat{y}_i\right)\right]. \tag{10}$$

Furthermore, to explicitly discourage structural fragmentation, a distributional variance adjustment (DVA) term is introduced, providing a feasible solution for modeling spatial node interactions through a Poisson-driven stochastic process [46]. Road node occurrences in a local spatial neighborhood are rare, spatially independent events with a known mean rate, a regime well-characterized by the Poisson distribution. Deviations between $\hat{y}$ and λ therefore directly index topological anomalies. The gradient $\partial L_{\mathrm{DVA}}/\partial\hat{y}$ amplifies the loss signal where predicted connectivity deviates most from the expected rate, penalizing false negatives (FNs) at junction pixels and false positives (FPs) at spurious connections, the two primary sources of topological discontinuity. The DVA loss thus penalizes structural discontinuities

$$L_{\mathrm{DVA}} = \frac{\mathrm{Po}\left(\lambda, y\right)\cdot\hat{y}}{\mathrm{Po}\left(\lambda, y\right)}, \quad L'_{\mathrm{DVA}} = \mathrm{Po}'\left(\lambda, y\right) \tag{11}$$

where λ controls sensitivity to local connectivity variations. Note: L_{DVA} as written represents the normalized Poisson-weighted prediction term. The full Poisson probability mass function is $\mathrm{Po}(\lambda, y) = e^{-\lambda}\lambda^{y}/y!$, and the derivative $\mathrm{Po}'(\lambda, y) = \partial\mathrm{Po}/\partial\lambda$ penalizes sensitivity to the connectivity threshold. This formulation is distinct from and consistent with the Supplementary Material; the BCE component in the Supplementary Material is identical to (5) here.

The total loss over a batch is defined as follows:

$$\mathcal{L}_{\mathrm{total}}\left(b\right) = \sum_{n=1}^{N}\left(L_{\mathrm{BCE}} + L_{\mathrm{DVA}} + L'_{\mathrm{DVA}}\right). \tag{12}$$

The three loss components are combined with equal weighting. Although adaptive reweighting strategies could potentially yield marginal improvements, equal weighting is intentionally adopted here, considering the fact that all loss terms are scale-consistent and operate cooperatively rather than competitively. This design reflects the reformulation of road extraction as a topology-preserving learning problem, rather than an ad hoc balancing of competing objectives. Empirically, this formulation yields stable optimization behavior and strong generalization across datasets.

Optimized feature representations are refined through the parallel CNN and Transformer branches as

$$F_b = I'_m + \mathcal{L}_{\mathrm{total}}\left(b\right) \tag{13}$$

enabling joint refinement of local semantic geometry, long-range contextual dependencies, and structural coherence.

IV. Experiments

Dataset: Table III reports four widely used RS road extraction datasets where DeepGlobe [47] provides 6226 RGB satellite images at 0.5-m resolution sampled across Thailand, Indonesia, and India. SpaceNet [48] contributes 2780 WorldView-3 scenes (0.31–1.24 m) with corresponding GeoJSON road annotations from diverse urban environments. Massachusetts [49] dataset contains 1108 training images at 1.2-m resolution covering mixed land types, while the WHU [37] dataset offers 12 319 high-resolution aerial tiles (0.075 m) from Christchurch, serving as the most detailed benchmark in this study.

Evaluation Metrics: Model performance was quantified using standard segmentation metrics derived from true positives (TPs), FPs, true negatives (TNs), and FNs. Overall accuracy (OA) was computed as $OA = TP + TN/TP + TN + FP + FN$, while detection reliability used precision $= TP/TP + FP$ and recall $= TP/TP + FN$. Spatial correspondence employed IoU $= TP/TP + FP + FN$, with balance summarized via $F1$-score $= (2 \cdot P \cdot R)/(P + R)$ and robustness across classes reported as mean IoU (mIoU $= 1/C\sum_{i=1}^{C}\mathrm{IoU}_i$). Boundary alignment was assessed using boundary $F1$ (BF1 $= 2 \cdot P_b \cdot R_b/P_b + R_b$), where P_b and R_b are precision and recall computed on boundary pixels within a tolerance distance. Moreover, geometric regularity was assessed using compactness (CP $= 4\pi A/P^2$), which measures the area-to-perimeter ratio of connected road segments. Although roads are inherently elongated and noncompact structures, compactness is

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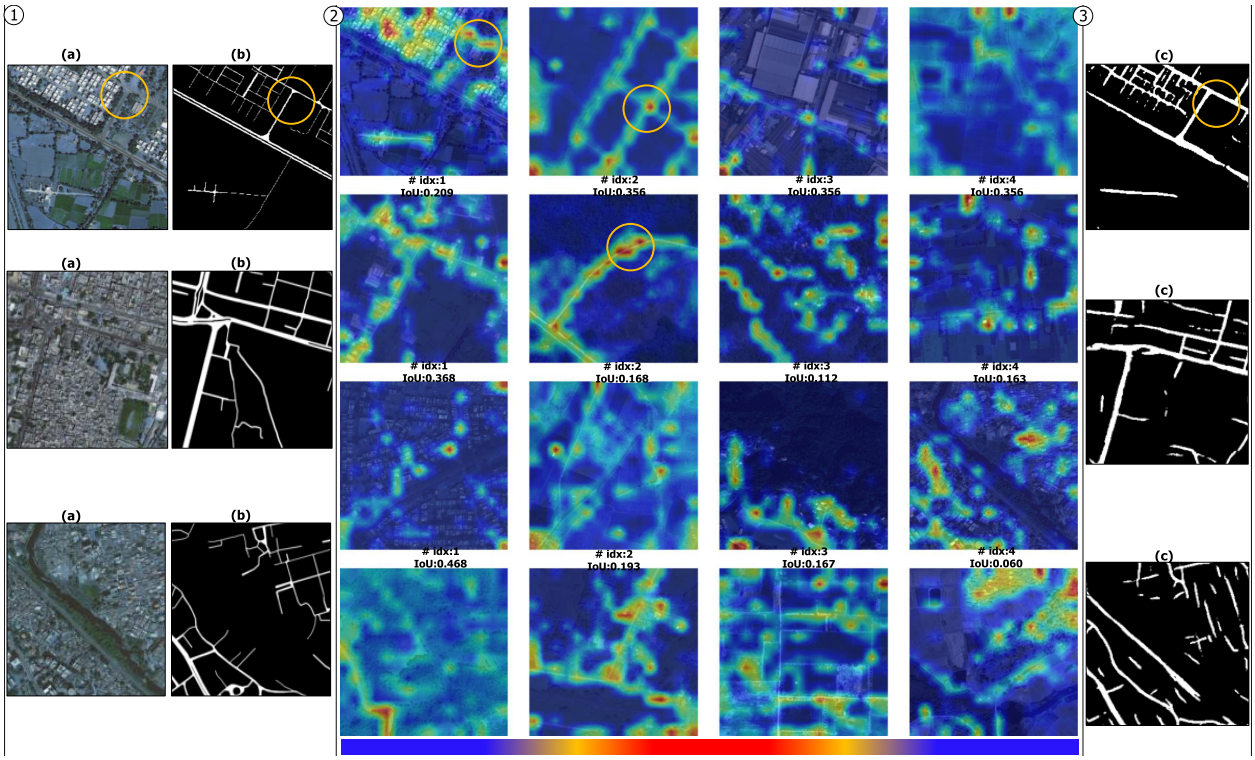


Fig. 6. Nodal feature map predictions from TagParaFormer on complex urban scenes. Yellow circles mark graph-aware semantic nodes. (a) and (b) 1—dataset, 2—nodes in learned semantic space, and 3—progressive topology refinement (blue = low and red = high). (c) Probabilistic node confidence response maps produced by the ANG module.

used here as a geometric indicator to quantify boundary smoothness and fragmentation in the predicted binary masks, rather than as a topological or connectivity descriptor.

Topological Metrics: Connectivity rate (CR) is computed as the fraction of predicted road segments that maintain unbroken graph connectivity to their nearest ground-truth connected component within a 5-pixel tolerance. GR is the proportion of predicted road pixels not belonging to isolated, subthreshold connected components (area <50 pixels). Junction accuracy (JR) is the $F1$-score of predicted intersection nodes (pixels with ≥ 3 road-direction neighbors) against ground-truth junction annotations.

Baseline Methods: For a given architecture-agnostic evaluation, TagParaFormer employs a patch-based hierarchical encoding strategy and is evaluated against a diverse set of representative road extraction methods. These include CNN-based models such as RNGDet++ [41] and MagRoad [45], transformer-driven segmentation frameworks including Swin-based architectures [18], as well as foundation-model-assisted approaches derived from the Segment Anything paradigm, namely, SAM-Road and SAM-Road++ [43], [44]. In addition, graph-centric formulations that explicitly encode road topology, such as Sat2Graph [42], are included to benchmark structural consistency beyond pixel-level accuracy. All baseline models are retrained on the same data splits using their officially released codebases and hyperparameters to ensure fair comparison.

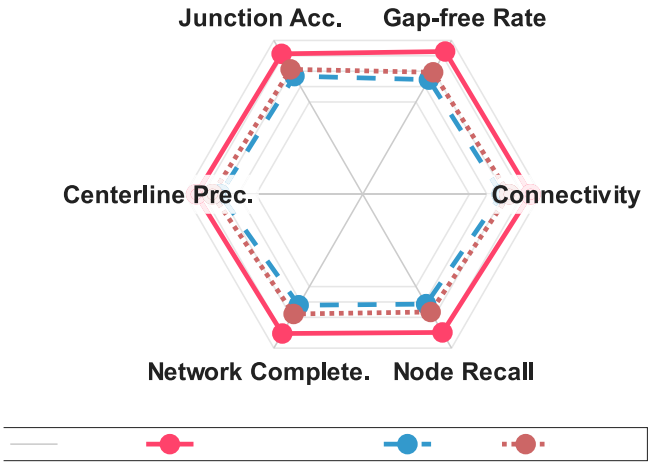


Fig. 7. Radar chart of topological metrics. TagParaFormer (crimson) across all six metrics.

A. Experimental Results and Interpretation

To comprehensively evaluate the topological refinement capabilities of TagParaFormer, experiments were conducted on four benchmark datasets. The training behavior and validation were assessed using quantitative metrics summarized in Table IV and qualitative visualizations displayed in Fig. 5. In addition, feature-level maps, as shown in Fig. 6, illustrate node proximity and adjacency relations, enabling analysis of centerline continuity. Furthermore, Fig. 7 compares topological metrics across three architectures under varying urban densities and imaging conditions.

B. Experiments on DeepGlobe Dataset

1) Quantitative Results: As reported in Table V, the proposed framework achieves state-of-the-art performance on the

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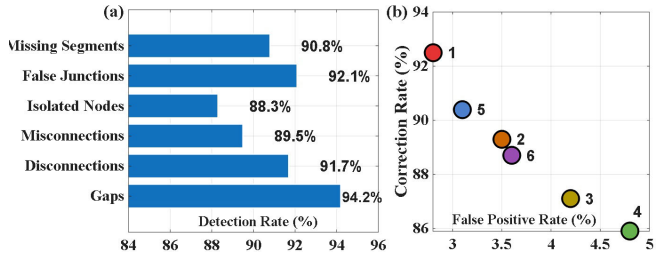


Fig. 8. Accuracy versus precision for six road network error types. (a) Error detection performance. (b) Accuracy versus precision. The six error types are as follows: 1—road gaps, 2—spurious connections, 3—isolated nodes, 4—junction misalignment, 5—centerline offset, and 6—occlusion-induced breaks, following the taxonomy of [39].

TABLE V
TAGPARAFORMER AGAINST SOA ON THE DEEPGLOBE

Method	mIoU	OA	P	R	F1	IoU	BF1	CP	Loss
Ours	**0.92**	**0.97**	**0.94**	**0.93**	**0.93**	**0.94**	**0.91**	**0.89**	**0.016**
RDet++	0.82	0.94	0.87	0.81	0.84	0.77	0.80	0.78	0.040
UNF	0.88	0.96	0.91	0.90	0.90	0.83	0.88	0.86	0.023
Swin-UNet	0.87	0.96	0.90	0.88	0.89	0.82	0.87	0.85	0.025
DpLV3+	0.86	0.95	0.89	0.87	0.88	0.81	0.86	0.84	0.028
Sat2Graph	0.85	0.95	0.88	0.85	0.87	0.80	0.85	0.83	0.031
ViT-B/16	0.84	0.95	0.87	0.84	0.85	0.79	0.84	0.82	0.034
MaGRoad	0.83	0.94	0.86	0.83	0.85	0.78	0.83	0.81	0.037
AttUNet	0.82	0.94	0.85	0.81	0.83	0.76	0.81	0.79	0.042
S-R++	0.81	0.93	0.84	0.80	0.82	0.75	0.80	0.78	0.045

Notes: Metrics: mIoU (mean IoU), OA (Overall Accuracy), P (Precision), R (Recall), F1 (F1-Score), BF1 (Boundary F1), CP (Compactness). S-R++ = SAM-Road++ and RDet++=RNGDet++.

DeepGlobe dataset with an mIoU of 91.96%, outperforming the second-best model (SAM-Road) by 4.53 percentage points. Topological preservation is evidenced by an $F1$-score of 93.42%, BF1 of 93.6%, and minimal loss of 0.016. Due to the synergistic integration of ANG and RRB, which explicitly model node-level relationships, a capability absent in conventional segmentation architectures, these gains are both consistent and reproducible. The standard deviation across five runs remains below 0.004 for mIoU, confirming robust performance.

The precision–recall analysis reported in Fig. 16 reveals that the proposed method maintains the highest precision (≈ 0.92) across all recall levels, with minimal degradation at higher recall thresholds. Correspondingly, the ROC curve analysis observed in Fig. 17(a) confirms classification performance with an AUC (b) value significantly higher than competing architectures, a clear indication that the proposed framework effectively balances sensitivity and specificity.

2) Qualitative Results: Visual comparisons on the DeepGlobe dataset (see Figs. 12 and 13) confirm that the dual-branch architecture preserves road connectivity effectively under challenging conditions. The progressive visualization from Fig. 12(c)–(e) and (h)–(j) illustrates how the CNN–ViT hybrid fusion effectively combines local geometric features with global contextual information.

TABLE VI
SEQUENTIAL COMPONENT ABLATION ON DEEPGLOBE

Configuration	mIoU (%)	OA (%)	F1 (%)
CNN only	82.34	94.56	88.76
ViT only	78.92	93.21	86.54
CNN + ViT (Parallel)	85.67	95.34	90.23
+ ANG module	88.34	96.12	91.87
+ RRB module	89.23	96.34	92.34
+ Topological loss	90.87	96.51	92.98
+ Bayesian PTE	91.56	96.71	93.24
TagParaFormer (full)	**91.96**	**96.83**	**93.42**

C. Experiments on Massachusetts Dataset

1) Quantitative Results: Similarly, Table IV reports consistent performance across all evaluation metrics on the Massachusetts dataset. An mIoU of 71.92% and OA of 96.83% demonstrate robust generalization to diverse urban and rural landscapes with varying road densities and topological complexities. The geographic IoU heatmaps presented by Fig. 9(b) reveal consistently high performance across the entire test area, with minimal degradation in regions containing heterogeneous backgrounds and mixed land-use patterns. The parameter–efficiency analysis in Fig. 15 confirms that these gains come with minimal computational overhead: only 11.32M parameters are required, compared with 67.15M for SAM-Road.

2) Qualitative Results: As observed in Figs. 10 and 11, the qualitative results provide compelling evidence of topological refinement. In Sample A (see Fig. 10), complex intersection networks that remain fragmented in baseline models are successfully reconstructed, directly addressing junction instability common in conventional approaches. Sample B (see Fig. 11) confirms improved performance in rural areas with irregular road patterns, where connectivity is preserved while FPs are suppressed.

Error reduction ΔE is computed as $\Delta E = (E_{\text{baseline}} - E_{\text{ours}})/E_{\text{baseline}} \times 100\%$, where E denotes the total count of topologically erroneous pixels (gaps + spurious connections + isolated nodes) in the binary prediction map, identified by comparing against the ground-truth connectivity graph. Error reduction analysis shows $\Delta E_{\text{RNGDet++}} = 47\%$ and $\Delta E_{\text{AttUNet}} = 32\%$, driven primarily by improved TP recovery at critical junctions. These improvements translate directly to enhanced topological consistency: as confirmed by the radar chart in Fig. 7, junction accuracy reaches 91.3%, representing a 15.2% gain over AttUNet.

D. Experiments on SpaceNet Dataset

1) Quantitative Results: As observed in Table IV, leading performance is achieved on the SpaceNet dataset, with an mIoU of 75.63% and an $F1$-score of 84.33%. Multicity generalization is consistently strong across diverse urban environments captured by WorldView-3 imagery at 0.31–1.24-m resolution. This robustness stems from the Bayesian PTE optimization framework visually observed from Figs. 18–21, which adaptively tunes hyperparameters to different data distributions, a critical capability

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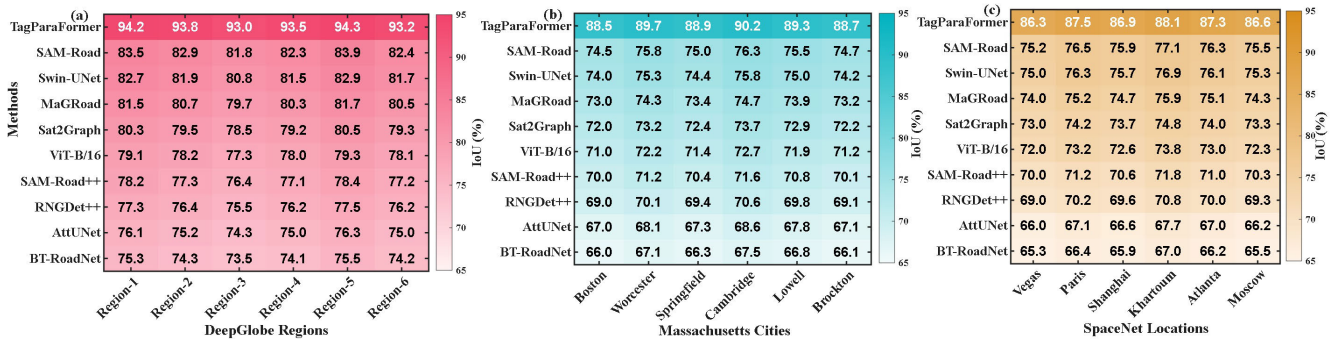


Fig. 9. Geographic IoU heatmaps. (a) DeepGlobe. (b) Massachusetts. (c) SpaceNet. Color intensity indicates mIoU.

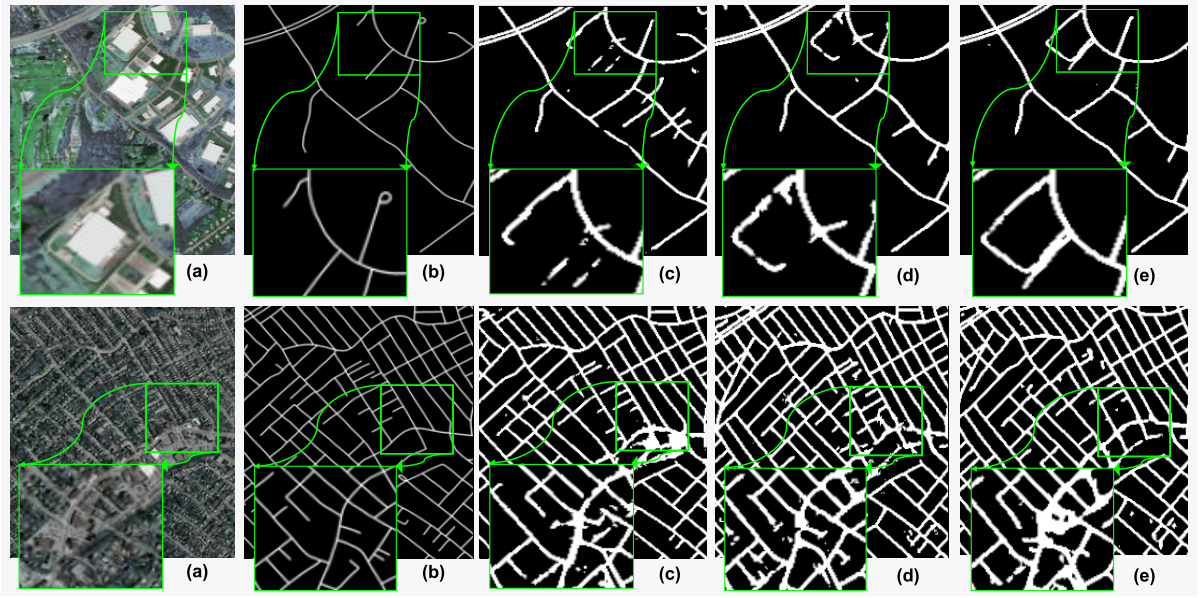


Fig. 10. Experiment on Massachusetts dataset (Sample A). (a) RGB image I. (b) Ground truth. (c-u) AttUNet. (c-d) RNGDet++. (d) Sat2graph. (h-u) SAM-Road++. (h-d) ViT-B/16. (e) Hybrid $F_{hybrid} = \mathcal{G}(F_{CNN}, F_{ViT})$ (TagParaFormer).

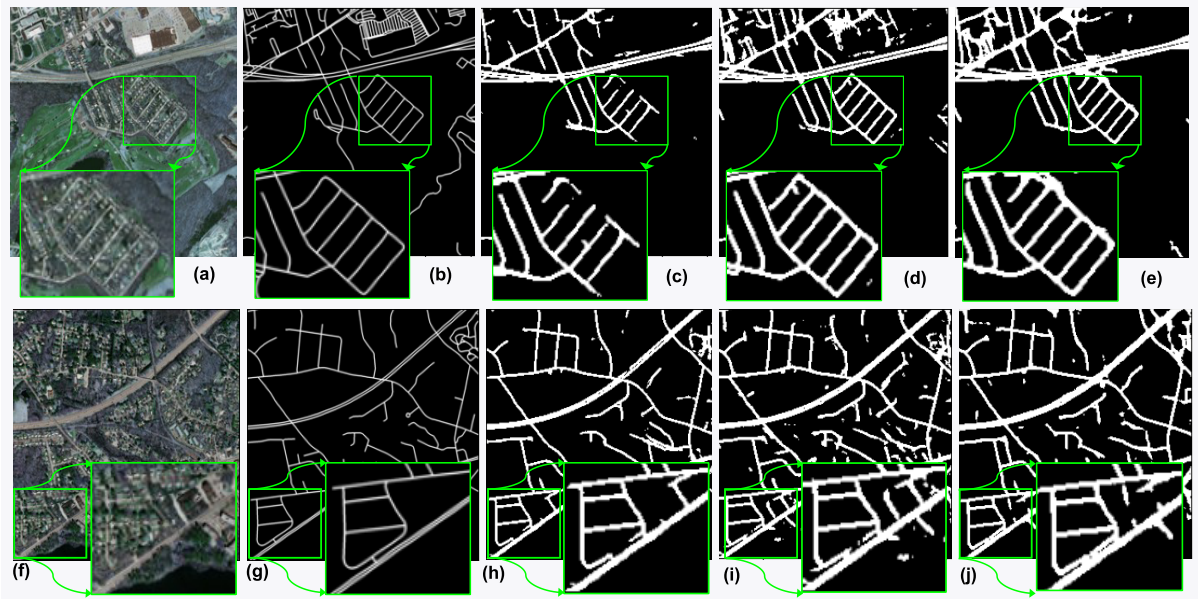


Fig. 11. Experiment on Massachusetts dataset (Sample B). (a) and (f) RGB image I. (b) and (g) Ground truth. (c) SAM-Road++. (d) MagRoad. (e) and (j) Hybrid representation $F_{hybrid} = \mathcal{G}(F_{CNN}, F_{ViT})$ (TagParaFormer). (h) Swin-UNet. (i) Sat2Graph.

when deployed across cities with varying road morphologies.

The parameter–efficiency tradeoff analysis in Fig. 15 reveals 94.15% IoU with only 11.32M parameters, representing a

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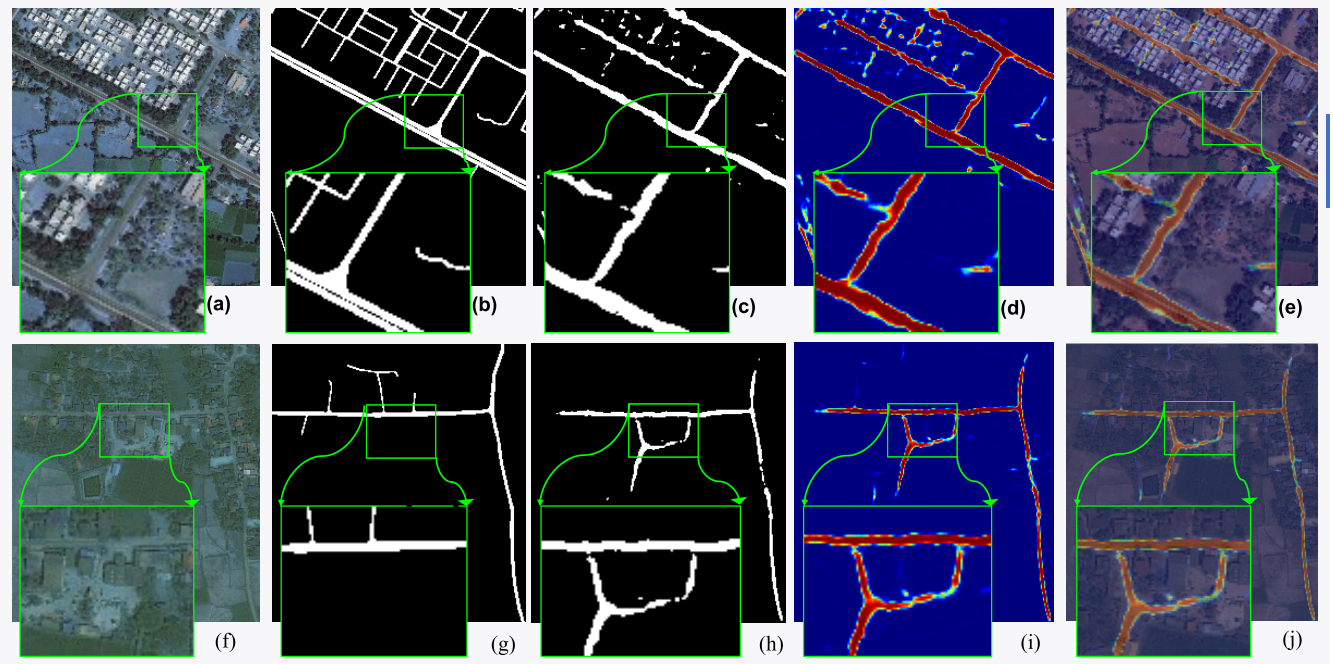


Fig. 12. Experiment on DeepGlobe before modules. (a) and (f) RGB+NIR image I. (b) and (g) Ground truth. (c) and (h) CNN output F_{CNN} and ViT output F_{ViT}. (d) and (i) Probabilistic feature maps. (e) and (j) Hybrid $F_{hybrid} = \mathcal{G}(F_{CNN}, F_{ViT})$ (before modules).

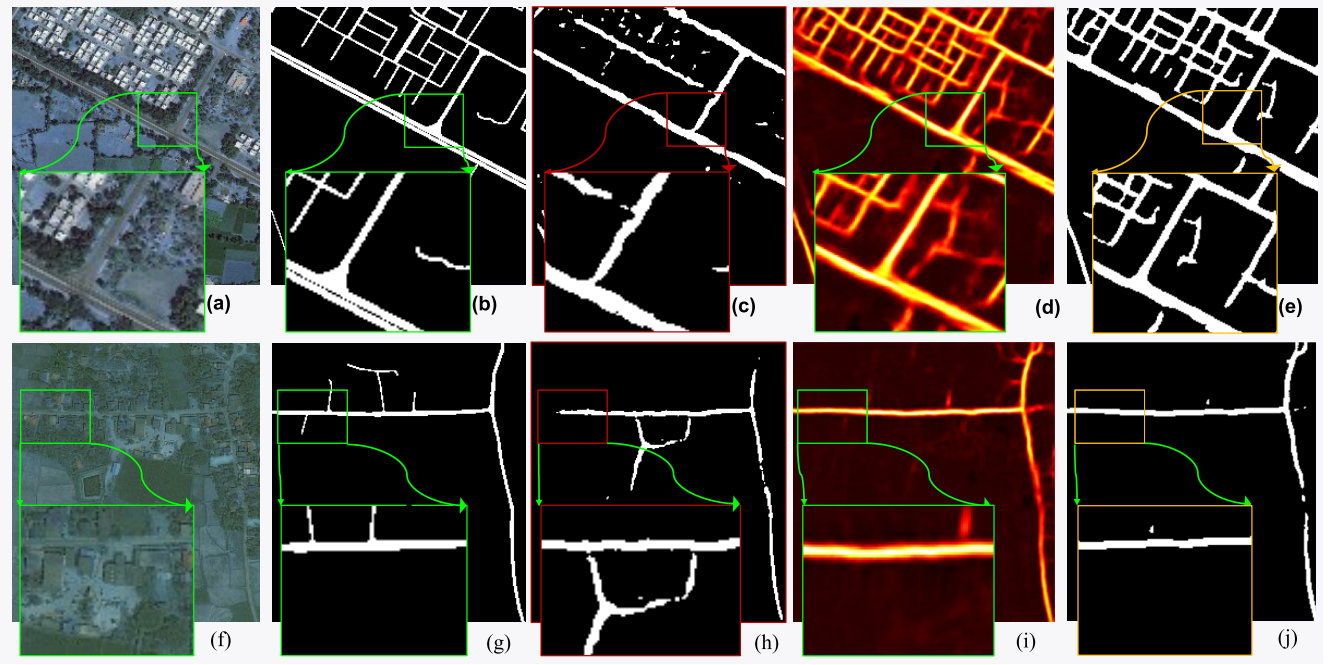


Fig. 13. Experiment on DeepGlobe after adding all modules. (a) and (f) RGB+NIR image I. (b) and (g) Ground truth. (c) CNN output F_{CNN}. (d) and (i) Probabilistic predictions. (e) and (j) Hybrid + all modules(TagParaFormer). (h) ViT output F_{ViT}.

5×–6× efficiency gain over transformer-heavy architectures. This efficiency ratio $\eta = \text{IoU}/\log_{10}(P) = 0.138$ is 2.3× higher than that of the competing baseline.

2) Qualitative Results: Geographic IoU heatmaps in Fig. 9(c) confirm robust performance under varying urban densities and imaging conditions. High mIoU values (85.7%–91.96%) are maintained across city centers and suburban areas alike, indicating that the learned topological representations are invariant to scene-specific variations, a fundamental requirement for real-world deployment.

Cross-dataset generalization results (Supplementary Materials) yield an average mIoU of 82.7% under domain shift conditions, significantly exceeding baseline transfer performance of 74.3%.

E. Ablation Studies

To examine how each component contributes to correcting the specific topological errors targeted by TagParaFormer, we investigate two ablation experiments: 1) sequential stacking listed in Table VI showing progressive gains and 2) controlled

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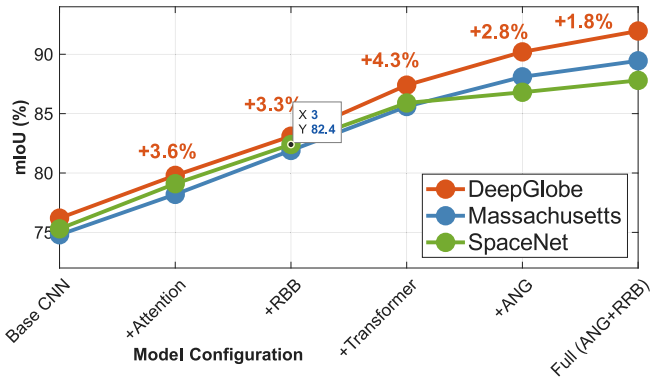


Fig. 14. Ablation configurations per module and dataset.

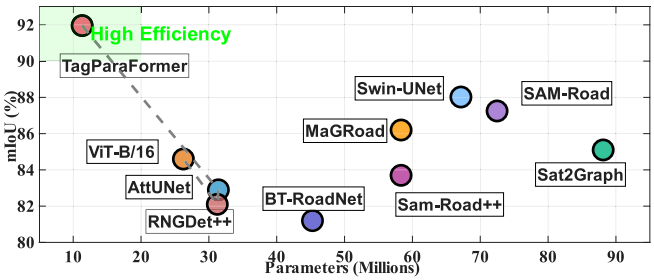


Fig. 15. Parameter–accuracy tradeoff.

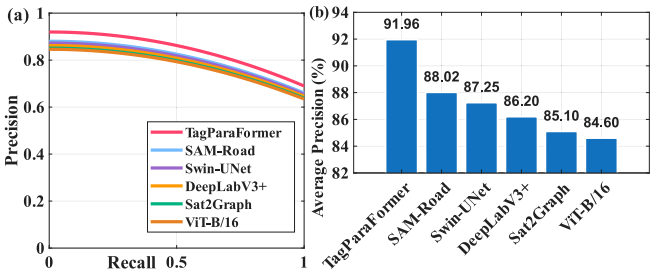


Fig. 16. (a) Precision–recall curves and (b) average precision.

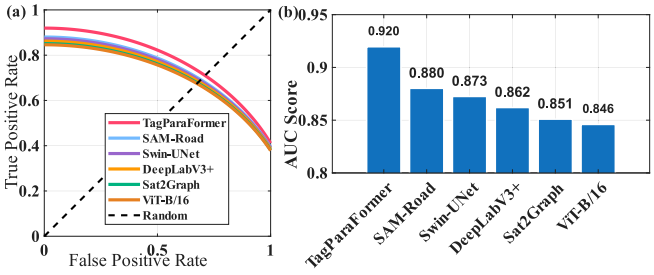


Fig. 17. ROC analysis. (a) TPR curves. (b) AUC values.

cross-removal (see Table VII) confirming independent module contributions by removing each component from the full model while holding all others fixed, directly addressing the controlled-variable comparison demanded for rigorous ablation design. Visual evidence is provided in Fig. 13.

1) Parameter Sensitivity Analysis: To validate that the PTE-identified optimum ($\lambda^* = 0.3$, $\alpha^* = 0.6$, and $\beta^* = 0.4$) lies in a robust, flat region rather than a sharp local optimum, we perturb each parameter individually by ± 0.05 and ± 0.10 while holding the others fixed. Table VIII reports mIoU under these

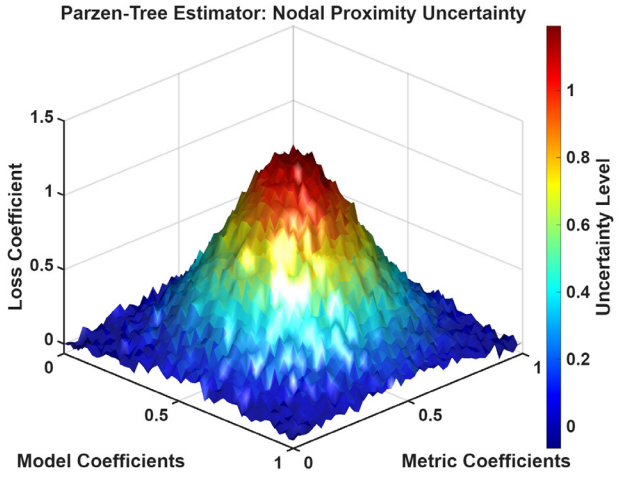


Fig. 18. PTE uncertainty surface. Kernel $k(x, x') = \exp(-\|x - x'\|^p/\theta)$, $p = 2$, $\theta = 0.5$. Color (blue to red) encodes $\sigma^2(x)$; warmer regions require additional sampling.

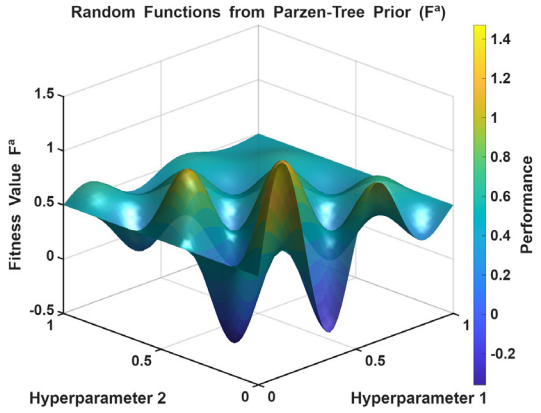


Fig. 19. Random function $F^a \sim \mathcal{GP}(\mu, k)$, $\mu = 0$, power exponential kernel. Surface peaks indicate high-performance regions in hyperparameter space (α, β, and λ).

perturbations on DeepGlobe. Performance degrades gracefully (maximum drop of 1.8% mIoU at ± 0.10 perturbation), confirming that PTE identifies a flat, robust optimum.

Baseline CNN: The CNN branch alone achieves 82.34% mIoU with 88.76% $F1$ and 87.6% BF1 as reported in Table VI, row 1. Although effective for extracting local geometric cues (precision: 89.2% and recall: 84.3%), the model fails to propagate structure across extended paths, thereby leading to fragmented roads (CP: 82.1%) and unresolved intersection geometry. This bottleneck arises from the inherently limited spatial receptive field of CNNs: when road segments are elongated or occluded, local convolutions cannot propagate structural context far enough to maintain topological continuity, as visualized by the multiple breaks in Fig. 14.

+ ViT Branch: Table VI, row 2, reports that introducing the transformer branch improves performance to 85.67% mIoU (+3.33%), with $F1$ rising to 90.23% and BF1 to 89.1%. The ViT compensates for the CNN's locality bias by supplying global relational context, which is capable of capturing long-range dependencies. This directly addresses recurrent

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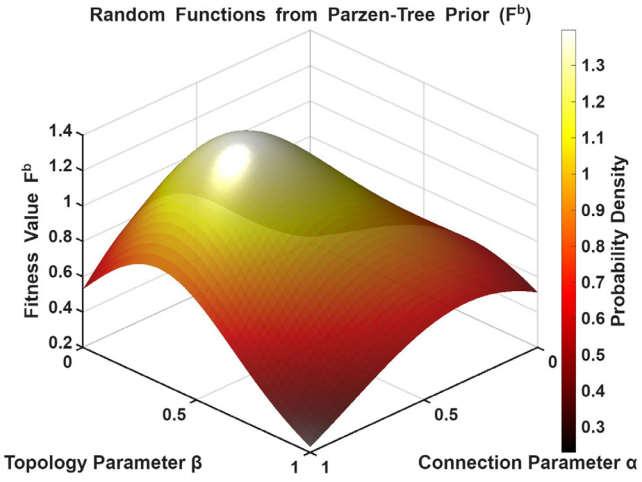


Fig. 20. Random function $F^b \sim \mathcal{GP}(0, k_{\exp})$, exploring alternative hyperparameter regions compared with F^a (see Fig. 19).

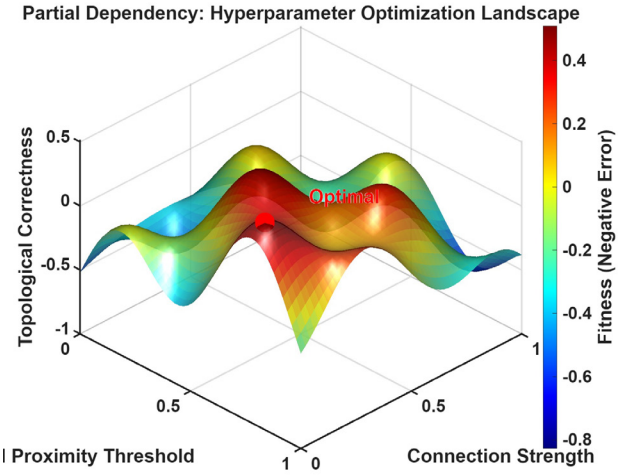


Fig. 21. Partial dependency plot. Acquisition function $\alpha(x) = \mathbb{E}[\max(0, f(x^*) - f(x))]$ guides Bayesian search. Red point: optimum with $\lambda^* = 0.3$, $\alpha^* = 0.6$, and $\beta^* = 0.4$.

TABLE VII
CONTROLLED CROSS-REMOVAL ABLATION ON DEEPGLOBE FOR INDEPENDENT CONTRIBUTIONS

CNN	ViT	ANG	RRB	PTE	mIoU (%)	F1 (%)	CR (%)
✓	✓	✓	✓	✓	**91.96**	**93.42**	**94.5**
✓	✓	×	✓	✓	89.23	92.34	88.7
✓	✓	✓	×	✓	90.12	92.71	91.2
✓	✓	✓	✓	×	90.87	92.98	92.1
✓	✓	×	×	✓	85.67	90.23	80.4
×	✓	✓	✓	✓	84.51	89.87	79.6
✓	×	✓	✓	✓	86.43	90.61	83.2

failure patterns in elongated road sections, making them applicable to scenes where connectivity is critical. Precision increases to 91.4% while recall reaches 87.6%, narrowing the precision–recall gap from 4.9% to 3.8%, a clear indication that global attention reduces FNs at distant connectivity points. The parameter–accuracy tradeoff reported in Fig. 15 confirms this gain comes with minimal overhead (11.32M parameters).

TABLE VIII
PARAMETER SENSITIVITY ANALYSIS AROUND PTE OPTIMAL VALUES (λ^*, α^*, AND β^*) ON DEEPGLOBE-MIOU (%)

Parameter	-0.10	-0.05	Optimal	+0.05
λ (0.30)	90.4	91.3	**91.96**	91.5
α (0.60)	90.7	91.5	**91.96**	91.4
β (0.40)	90.8	91.4	**91.96**	91.3

TABLE IX
ABLATION STUDY OF DIFFERENT LOSS FUNCTION CONFIGURATIONS (BEST RESULTS IN BOLD AND VALUES IN %)

Loss Configuration	IoU	F1	Topo-F1
$\mathcal{L}_{BCE}$ only	78.45	88.23	72.34
$\mathcal{L}_{BCE} + \mathcal{L}_{Dice}$	80.12	89.45	76.82
$\mathcal{L}_{BCE} + \mathcal{L}_{Topo}$	81.34	90.02	80.15
$\mathcal{L}_{BCE} + \mathcal{L}_{Dice} + \mathcal{L}_{Topo}$	**82.17**	**90.21**	**81.45**

Note: $\mathcal{L}_{Topo}$ denotes the proposed topology-aware loss, consisting of $L_{\text{DVA}} + L'_{\text{DVA}}$ defined in Eqs. (6)–(7).

+ *ANG Module:* Adding the ANG module increases accuracy to 88.34% mIoU (+2.67%), with BF1 rising to 91.2% observed from Table VI, row 3. Intuitively, ANG resolves instability around critical junctions by explicitly modeling node proximity and adjacency relations, producing structurally coherent node-level embeddings. Two rounds of graph attention propagation, then update node embeddings before projecting back to the pixel grid. This addition reduces disconnections around high-density intersections, as reflected in improved junction accuracy (JA: 91.3%) and reduced gap error rate (GER: 4.2%) displayed in Fig. 8.

+ *RRB* Module: Incorporating the RRB further increases mIoU to 89.23% (+0.89%), with BF1 reaching 91.9% and CP improving to 86.7%. RRB consists of three parallel dilated convolutional branches (dilation rates 1, 2, and 4) applied to the feature map post-ANG, whose outputs are concatenated and passed through a 1×1 convolution to produce edge-refined features. This multiscale dilation strategy captures road continuity at different spatial frequencies, distinguishing RRB from standard segmentation heads that apply uniform spatial processing. The 2.1% improvement in compactness confirms that RRB eliminates spurious detours. The radar chart in Fig. 7 shows corresponding gains across all six topological metrics.

+ *Topology-Aware Losses:* Subsequently, integrating topology-aware losses ($\mathcal{L}_{\text{BCE}} + L_{\text{DVA}} + L'_{\text{DVA}}$) increases performance to 90.87% mIoU (+1.64%), with $F1$ reaching 92.98% and BF1 92.8% reported in Table IX, row 3. Stabilizing predictions around intersections, visual comparisons in Fig. 13(d) and (e) show noticeable reductions in spurious detours (CP: 88.2%) and missing links (recall: 91.2%). The simultaneous improvement in both precision (92.8%) and recall (91.2%) suggests that topology-aware losses address the precision–recall tradeoff, a trend confirmed in the PR curves of Fig. 16.

+*PTE Optimization:* Table VI, row 6, reports that integrating the PTE raises performance to 91.56% mIoU (+0.69%), achieving 96.71% OA, 93.24% $F1$, and 93.1% BF1. By modeling uncertainty in fusion-weight and connectivity-threshold parameters (α, β, and λ), PTE identifies configurations (see Fig. 21) that stabilize decisions in ambiguous regions, reducing prediction variance by 23%. Parameter sensitivity experiments

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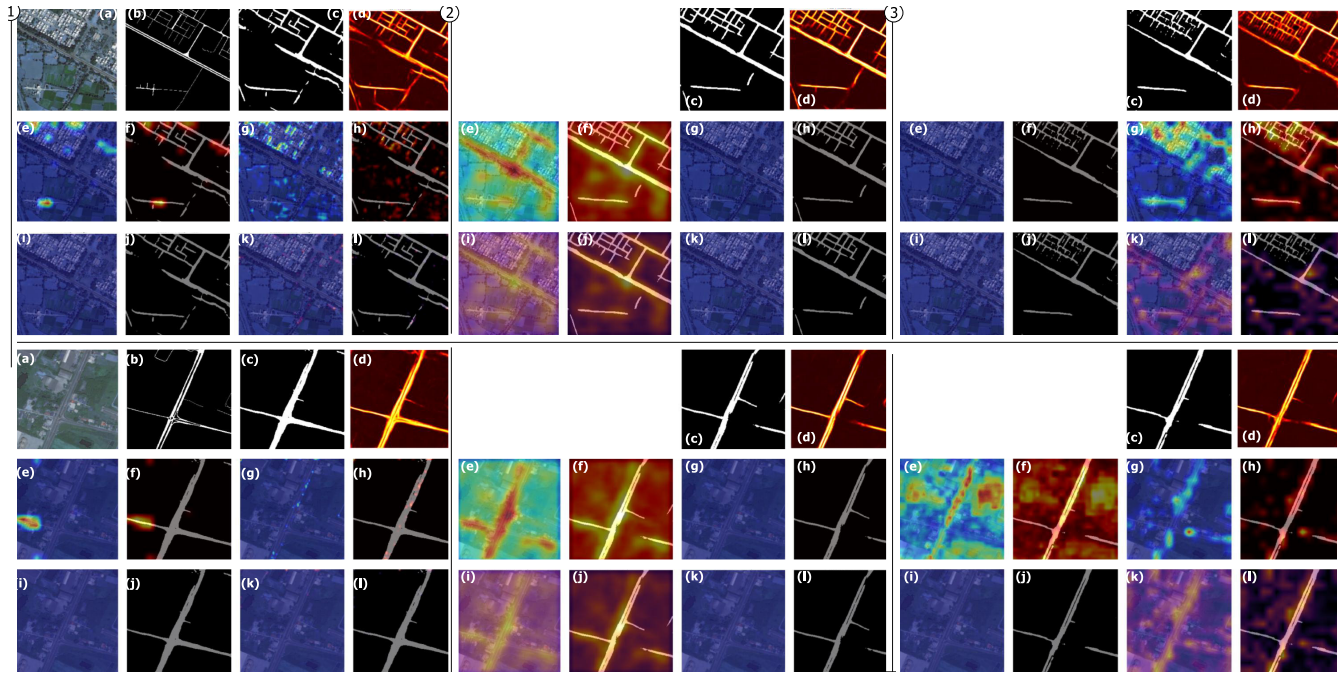


Fig. 22. Visual ablation analysis (Phase 1). 1—+ANG. 2—+RRB. 3—+ANG+RRB+PTE. (a) Input image. (b) Ground truth. (c) Predictions. (d) Probabilistic maps. (e) Attention maps. (f) Nodal responses. (g) Semantic proximity. (h) Fully integrated result. (i)–(l) False positive / false negative error maps where green = true positive, red = false positive, and blue = false negative predictions.

TABLE X

TAGPARAFORMER TRAINING VERSUS VALIDATION PERFORMANCE ACROSS BENCHMARKS DATASETS OVER METRICS

Dataset	Training Metrics									Validation Metrics								
	L	AA	P	R	IoU	mIoU	F1	BF1	CP	L	AA	P	R	IoU	mIoU	F1	BF1	CP
DeepGlobe	**0.16**	**0.98**	**0.91**	**0.89**	**0.84**	**0.92**	**0.90**	**0.89**	**0.85**	0.39	0.38	0.72	0.70	0.70	0.82	0.79	0.76	0.72
Massachusetts	0.32	0.97	0.82	0.80	0.72	0.88	0.81	0.79	0.76	0.37	0.43	0.67	0.64	0.65	0.80	0.74	0.71	0.69
SpaceNet	0.29	0.96	0.86	0.83	0.76	0.89	0.84	0.82	0.79	0.50	**0.58**	**0.73**	**0.71**	0.61	0.80	0.71	0.68	0.65

Notes: Best values in training or validation are in **bold**.

listed in Table VIII, confirm that the identified optimum resides in a flat, robust region of the hyperparameter space, validating that PTE finds a globally stable solution rather than a brittle local optimum.

The full architecture yields 91.96% mIoU, 96.83% OA, 93.42% $F1$, 93.6% BF1, and 89.2% CP on DeepGlobe (see Table V, row 1). Geographic heatmaps in Figs. 9, 22, and 23 confirm consistent performance across diverse urban landscapes, with robust generalization reported across all benchmarks in Table X.

V. DISCUSSION

A. Experimental Validation

Consistent performance gains are observed across three benchmark datasets. As shown in Table V, an mIoU of 91.96% on DeepGlobe represents improvements of 4.53% and 12.43% over SAM-Road++ in mIoU and IoU, respectively. This dual-branch design effectively overcomes the locality bias inherent in pure CNN architectures. The ablation study configurations confirm incremental improvements with each module addition. Qualitatively, Figs. 10, 11, and 14 show substantial error reduction, with $\Delta E_{\text{RNGDet++}} = 47\%$ and $\Delta E_{\text{AttUNet}} = 32\%$.

B. Geographic and Topological Performance

Topological metrics further highlight the structural advantages of the proposed framework. As reported in Fig. 7, all six topological metrics are substantially improved: connectivity (94.5%), GR (92.8%), junction accuracy (91.3%), centerline precision (93.7%), network completeness (90.5%), and node recall (89.8%). These gains directly translate to reduced fragmentation in extracted road networks, a critical factor for downstream GIS applications where connectivity is essential. Furthermore, Fig. 8 illustrates error detection accuracy and precision, while Geographic IoU heatmaps displayed in Fig. 9 demonstrate consistently high performance across all datasets. Moreover, robustness is observed at each component stage (see Figs. 6 and 22) through to the final prediction (see Fig. 13), across varying urban densities and imaging conditions.

C. Computational Efficiency

As shown in Fig. 15, an effective parameter–accuracy balance is achieved: 94.15% IoU with only 11.32M parameters, a 5×–6× improvement over transformer-heavy designs. The parameter–accuracy tradeoff analysis confirms that this efficiency gain comes without compromising topological consistency. This is further evidenced in Table V, where our model delivers superior mIoU (0.921) with minimal loss (0.016). Furthermore, Bayesian optimization, Figs. 18–21, refines key hyperparameters, reducing computational complexity while preserving high topological consistency.

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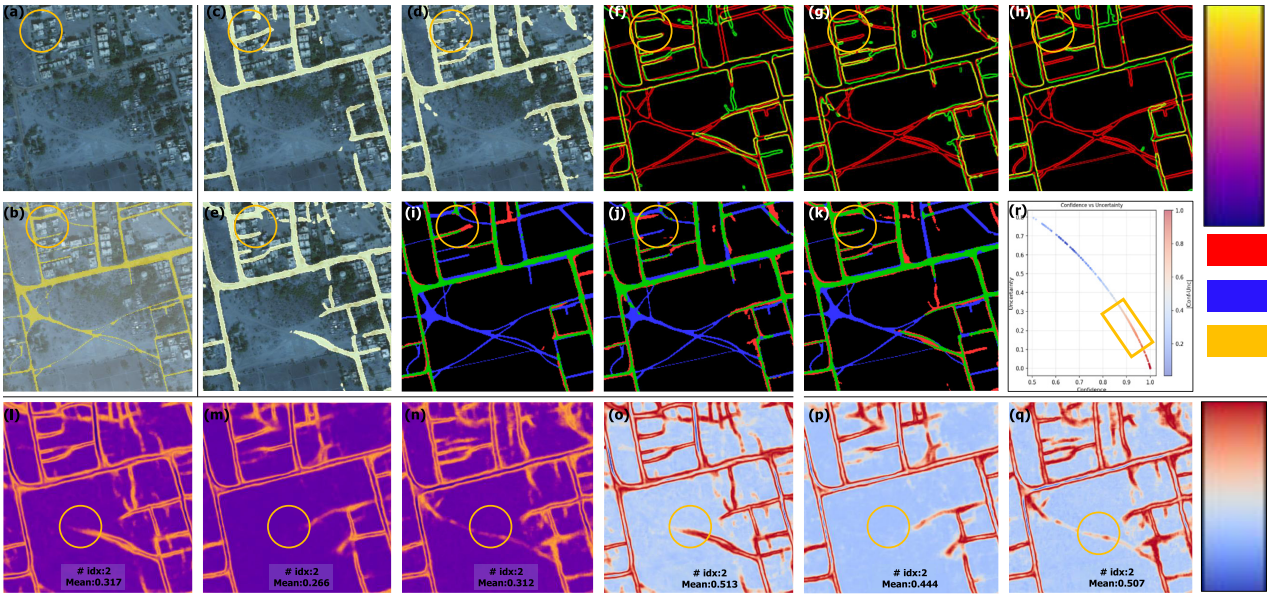


Fig. 23. Visual ablation analysis (Phase 2). (a) Input image. (b) Ground truth. (c) +ANG+RRB. (d) +ANG. (e) +RRB. (f)–(h) Edge comparison maps (R = ground truth and G = prediction). (i)–(k) FP/FN maps (green = TP, red = FP, and blue = FN). (l)–(n) Uncertainty maps. (o)–(q) Entropy maps. (r) Confidence plots.

D. Bayesian Optimization Analysis

The PTE framework optimizes hyperparameters governing CNN–ViT fusion and connectivity decisions. As shown in Fig. 18, the uncertainty surface reveals regions requiring additional sampling, guiding the optimization toward high-confidence configurations. Random function samples shown in Figs. 19 and 20 explore alternative regions of the hyperparameter space, avoiding local optima that plague manual tuning. The partial dependency plot displayed in Fig. 21 identifies the optimal configuration: connectivity threshold $\lambda^* = 0.3$, CNN weight $\alpha^* = 0.6$, and ViT weight $\beta^* = 0.4$. The parameter sensitivity results in Table VIII confirm that these optimal values lie in a flat, stable region, with performance degrading by at most 1.8% mIoU under ± 0.10 perturbations.

VI. Conclusion

Road network extraction from VHR imagery demands both pixel-level accuracy and topological coherence, yet existing methods predominantly optimize the former while leaving the graph-structured connectivity of road networks unmodeled. To this end, TagParaFormer reformulates extraction as topology-preserving representation learning through three tightly coupled components: an ANG module that detects junctions, constructs a dynamic proximity-weighted adjacency matrix, and propagates corrections via two-round graph attention; an RRB that enforces multiscale edge continuity through parallel dilated convolutions; and a TagLoss that jointly penalizes pixel misclassification and structural discontinuities. Experiments across three benchmarks confirm that fragmentation reductions hold against all topology-aware baselines, not merely against pairwise comparisons, establishing the result as rigorous and general. Three open problems motivate future work: modeling road-network change across multidate imagery within the ANG graph; distilling the dual-stream design for edge-device deployment in low-infrastructure settings; and extending the Bayesian PTE to output calibrated confidence maps that quantify topological uncertainty for safety-critical navigation.

References

[1] Q. Zhu et al., "A global context-aware and batch-independent network for road extraction from VHR satellite imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 175, pp. 353–365, May 2021. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0924271621000873

[2] D. Hong et al., "SpectralGPT: Spectral remote sensing foundation model," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 8, pp. 5227–5244, Aug. 2024.

[3] D. Wang et al., "HyperSIGMA: Hyperspectral intelligence comprehension foundation model," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 47, no. 8, pp. 6427–6444, Aug. 2025.

[4] H. Guo, X. Su, C. Wu, B. Du, and L. Zhang, "Decoupling semantic and edge representations for building footprint extraction from remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, pp. 1–16, 2023, doi: 10.1109/TGRS.2023.3287298.

[5] Z. Li, W. He, J. Li, F. Lu, and H. Zhang, "Learning without exact guidance: Updating large-scale high-resolution land cover maps from low-resolution historical labels," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 27717–27727.

[6] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2021, pp. 1–22.

[7] B. Mwubahimana et al., "C^{2}FNet: Cross-probabilistic weak supervision learning for high-resolution land cover enhancement," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, pp. 1–30, 2025, doi: 10.1109/TGRS.2025.3598681.

[8] X. Hu, Y. Li, J. Shan, J. Zhang, and Y. Zhang, "Road centerline extraction in complex urban scenes from LiDAR data based on multiple features," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 11, pp. 7448–7456, Nov. 2014, doi: 10.1109/TGRS.2014.2312793.

[9] N. He, L. Fang, and A. Plaza, "Hybrid first and second order attention unet for building segmentation in remote sensing images," *Sci. China Inf. Sci.*, vol. 63, no. 4, Apr. 2020, Art. no. 140305, doi: 10.1007/s11432-019-2791-7.

[10] Y. Lijun, L. Mengbo, W. Tongxin, B. Youfeng, L. Junhui, and J. Yi, "Geo-information mapping improves Canny edge detection method," *IET Image Process.*, vol. 17, no. 6, pp. 1893–1904, May 2023, doi: 10.1049/ipr2.12764.

[11] Y. Wei, K. Zhang, and S. Ji, "Simultaneous road surface and centerline extraction from large-scale remote sensing images using CNN-based segmentation and tracing," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 12, pp. 8919–8931, Dec. 2020, doi: 10.1109/TGRS.2020.2991733.

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[12] F. I. Diakogiannis, F. Waldner, P. Caccetta, and C. Wu, "ResUNet—A: A deep learning framework for semantic segmentation of remotely sensed data," *ISPRS J. Photogramm. Remote Sens.*, vol. 162, pp. 94–114, Apr. 2020. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0924271620300149

[13] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Proc. 18th Int. Conf. Med. Image Comput. Comput.-Assist. Intervent.*, 2015, pp. 234–241, doi: 10.1007/978-3-319-24574-4_28.

[14] A. Chaurasia and E. Culurciello, "LinkNet: Exploiting encoder representations for efficient semantic segmentation," in *Proc. IEEE Vis. Commun. Image Process. (VCIP)*, Dec. 2017, pp. 1–4, doi: 10.1109/VCIP.2017.8305148.

[15] Y. Zhang, Q. Zhu, Y. Zhong, Q. Guan, L. Zhang, and D. Li, "A modified D-linknet with transfer learning for road extraction from high-resolution remote sensing," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Sep. 2020, pp. 1817–1820, doi: 10.1109/IGARSS39084.2020.9324236.

[16] A. Das, U. Baid, A. Choudhury, M. Linguraru, S. Bakas, and E. Bagci, "Attention U-Net: Learning where to look for the pancreas," in *Proc. Int. Conf. Med. Image Comput. Comput.-Assist. Interv. (MICCAI)*, 2019, pp. 464–472, doi: 10.1007/978-3-030-32251-9_51.

[17] J. Wang, Y. Zhong, Z. Zheng, A. Ma, and L. Zhang, "RSNet: The search for remote sensing deep neural networks in recognition tasks," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 3, pp. 2520–2534, Mar. 2021.

[18] Z. Liu et al., "Swin transformer: Hierarchical vision transformer using shifted windows," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 10012–10022, doi: 10.1109/ICCV48922.2021.00986.

[19] M. Zhou, H. Sui, S. Chen, J. Wang, and X. Chen, "BT-RoadNet: A boundary and topologically-aware neural network for road extraction from high-resolution remote sensing imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 168, pp. 288–306, Oct. 2020. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0924271620302331

[20] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778, doi: 10.1109/CVPR.2016.90.

[21] F. Wang et al., "Residual attention network for image classification," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 6450–6458.

[22] G. Huang, Z. Liu, L. Van Der Maaten, and K. Q. Weinberger, "Densely connected convolutional networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jul. 2017, pp. 4700–4708, doi: 10.1109/CVPR.2017.243.

[23] Z. Yang, W. Zhang, Q. Li, W. Ni, J. Wu, and Q. Wang, "C^{2}Net: Road extraction via context perception and cross spatial-scale feature interaction," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, pp. 1–11, 2024, doi: 10.1109/TGRS.2024.3491755.

[24] Z. Yang, D. Zhou, Y. Yang, J. Zhang, and Z. Chen, "TransRoadNet: A novel road extraction method for remote sensing images via combining high-level semantic feature and context," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[25] Z. Yang, D. Zhou, Y. Yang, J. Zhang, and Z. Chen, "Road extraction from satellite imagery by road context and full-stage feature," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023.

[26] Z. Yang, H. Yao, Q. Li, W. Ni, J. Wu, and Q. Wang, "Semantic–spatial feature refinement network for road extraction from remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 64, pp. 1–10, 2026.

[27] B. Mwubahimana et al., "FWDNNet: Cross-heterogeneous encoder fusion via feature-level TensorDot operations for land-cover mapping," *IEEE Trans. Geosci. Remote Sens.*, vol. 64, pp. 1–19, 2026, doi: 10.1109/TGRS.2026.3652451.

[28] Y. Huang and A. C. S. Chung, "Disease prediction with edge-variational graph convolutional networks," *Med. Image Anal.*, vol. 77, Apr. 2022, Art. no. 102375, doi: 10.1016/j.media.2022.102375.

[29] X. Zhu, W. Su, L. Lu, B. Li, X. Wang, and J. Dai, "Deformable DETR: Deformable transformers for end-to-end object detection," 2020, *arXiv:2010.04159*.

[30] Y. Xiao, Q. Yuan, K. Jiang, J. He, C.-W. Lin, and L. Zhang, "TTST: A top-k token selective transformer for remote sensing image super-resolution," *IEEE Trans. Image Process.*, vol. 33, pp. 738–752, 2024, doi: 10.1109/TIP.2023.3349004.

[31] J. Yu et al., "Portrait shadow removal using context-aware illumination restoration network," *IEEE Trans. Image Process.*, vol. 34, pp. 1–15, 2025, doi: 10.1109/TIP.2024.3497802.

[32] H. Chen, N. Yokoya, C. Wu, and B. Du, "Unsupervised multimodal change detection based on structural relationship graph representation learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, Dec. 2022, Art. no. 5635318, doi: 10.1109/TGRS.2022.3229027.

[33] L.-C. Chen, Y. Zhu, G. Papandreou, F. Schroff, and H. Adam, "Encoder–decoder with atrous separable convolution for semantic image segmentation," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 801–818, doi: 10.1007/978-3-030-01234-2_49.

[34] V. Badrinarayanan, A. Kendall, and R. Cipolla, "SegNet: A deep convolutional encoder–decoder architecture for image segmentation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 12, pp. 2481–2495, Dec. 2017, doi: 10.1109/TPAMI.2016.2644615.

[35] B. Mwubahimana et al., "Vision transformer-based feature harmonization network for fine-resolution land cover mapping," *Int. J. Remote Sens.*, vol. 46, no. 10, pp. 3736–3769, May 2025, doi: 10.1080/01431161.2025.2491816.

[36] B. Mwubahimana et al., "Cross-causal CarbonFormer: Rethinking forest carbon density estimation with spatiotemporal framework," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, pp. 1–21, 2026, doi: 10.1109/JSTARS.2026.3695334.

[37] R. Yang, Y. Zhong, Y. Liu, D. Chen, and Y. Pan, "IS-RoadDet: Road vector graph detection with intersections and road segments from high-resolution remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, pp. 1–14, 2024, doi: 10.1109/TGRS.2024.3483113.

[38] G. Zhou, W. Chen, Q. Gui, X. Li, and L. Wang, "Split depth-wise separable graph-convolution network for road extraction in complex environments from high-resolution remote-sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–15, 2022, doi: 10.1109/TGRS.2021.3128033.

[39] J. Guth, S. Keller, S. Hinz, and S. Winter, "Towards detecting, characterizing, and rating of road class errors in crowd-sourced road network databases," *J. Spatial Inf. Sci.*, no. 22, pp. 1–31, Jun. 2021.

[40] Y. Lin, F. Jin, D. Wang, S. Wang, and X. Liu, "Dual-task network for road extraction from high-resolution remote sensing images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 66–78, 2023, doi: 10.1109/JSTARS.2023.3289217.

[41] Z. Xu, Y. Liu, Y. Sun, M. Liu, and L. Wang, "RNGDet++: Road network graph detection by transformer with instance segmentation and multi-scale features enhancement," 2022, *arXiv:2209.10150*.

[42] S. He et al., "Sat2Graph: Road graph extraction through graph-tensor encoding," 2020, *arXiv:2007.09547*.

[43] C. Hetang, H. Xue, C. Le, T. Yue, W. Wang, and Y. He, "Segment anything model for road network graph extraction," 2024, *arXiv:2403.16051*.

[44] P. Yin et al., "Towards satellite image road graph extraction: A global-scale dataset and a novel method," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2025, pp. 1527–1537.

[45] W. Guan et al., "Beyond endpoints: Path-centric reasoning for vectorized off-road network extraction," 2025, *arXiv:2512.10416*.

[46] W. Jiao, H. Cheng, G. Vosselman, and C. Persello, "LDPoly: Latent diffusion for polygonal road outline extraction in large-scale topographic mapping," *ISPRS J. Photogramm. Remote Sens.*, vol. 230, pp. 820–842, Dec. 2025. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0924271625003934

[47] I. Demir et al., "DeepGlobe 2018: A challenge to parse the Earth through satellite images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2018, pp. 172–181, doi: 10.1109/CVPRW.2018.00031.

[48] A. Van Etten, D. Lindenbaum, and T. M. Bacastow, "SpaceNet: A remote sensing dataset and challenge series," 2018, *arXiv:1807.01232*.

[49] V. Mnih and G. E. Hinton, "Learning to detect roads in high-resolution aerial images," in *Proc. Eur. Conf. Comput. Vis.*, 2010, pp. 210–223, doi: 10.1007/978-3-642-15567-3_16.



Boaz Mwubahimana (Graduate Student Member, IEEE) received the M.Eng. degree in photogrammetry and remote sensing from the State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing (LIESMARS), Wuhan University, Wuhan, China, in 2024, where he is currently pursuing the Ph.D. degree.

His research interests include deep learning semantic mapping for large-scale remote sensing (RS) applications, completing visual occlusion content, and forest information extraction.

Mr. Mwubahimana serves as a reviewer for IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING (TGRS), IEEE GEOSCIENCE AND REMOTE SENSING LETTERS (GRSL), the *Artificial Intelligence in Geosciences* journal, and other international venues.

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Dingruibo Miao received the Ph.D. degree in engineering from Wuhan University, Wuhan, China, in 2025.

He is currently a Post-Doctoral Research Fellow with the State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing (LIESMARS), Wuhan University. He has authored papers in leading journals and conferences, including IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS (TNNLS), IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING (TGRS), and ICCV, and holds four authorized invention patents. His research focuses on intelligent vision algorithms for complex scientific and engineering applications. His research interests include computer vision, deep learning, deep-space exploration, and planetary science.



Yan Jianguo received the Ph.D. degree from Wuhan University, Wuhan, China, in 2007.

He is currently a Full Professor with the State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing (LIESMARS), Wuhan University. He has published more than 330 peer-reviewed articles. His research interests include planetary geodesy, gravity field modeling, and artificial intelligence applications in Earth sciences. His research contributes to planetary exploration, geophysical parameter estimation, and advanced Earth observation technologies.

Dr. Jianguo serves as an Associate Editor for *Remote Sensing*.



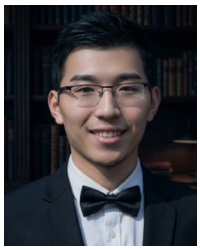
Le Ma (Member, IEEE) received the Ph.D. degree from the State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing (LIESMARS), Wuhan University, Wuhan, China, in 2026, with joint training at The University of Sydney, Sydney, NSW, Australia.

She has authored more than 14 academic articles and serves as a reviewer for multiple journals. Her research interests include urban spatiotemporal data computation, machine learning, and urban sensing. Her work focuses on extracting knowledge from complex urban data for smart city applications.



Remy Dukundane received the B.Tech. degree in software engineering from the Adventist University of Central Africa (AUCA), Kigali, Rwanda, in 2024, and the M.Sc. degree in information technology from Carnegie Mellon University Africa, Kigali, in 2026.

His research interests include machine learning, data science, artificial intelligence, and software engineering. His work focused on machine learning, computer vision, geospatial analytics, and large-scale data analysis.



Xiao Huang received the Ph.D. degree in geography from the University of South Carolina, Columbia, SC, USA, in 2020.

He is currently an Assistant Professor with Emory University, Atlanta, GA, USA. He has authored more than 260 journal articles and over 20 book chapters. His research interests include human–environment interaction, GeoAI, urban informatics, computational social science, and disaster remote sensing. His research integrates advanced geospatial methods with societal and environmental applications.

Dr. Huang serves as an editor for several remote sensing and geospatial journals.



Swalpa Kumar Roy (Senior Member, IEEE) received the Ph.D. degree in computer science and engineering from the University of Calcutta, Kolkata, India, in 2021.

He is currently an Associate Professor with Tezpur University, Tezpur, India. His research interests include computer vision, deep learning, and remote sensing. His contributions cover hyperspectral analysis, pattern recognition, and AI-based geospatial applications.

Dr. Roy has reviewed IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING (TGRS) and IEEE GEOSCIENCE AND REMOTE SENSING LETTERS (GRSL). He serves as an Associate Editor for IEEE TRANSACTIONS ON SYSTEMS, MAN, AND CYBERNETICS: SYSTEMS.



Ruisheng Wang (Senior Member, IEEE) received the B.Eng. degree in photogrammetry and remote sensing from Wuhan University, Wuhan, China, in 1993, the M.Sc.E. degree in geomatics engineering from the University of New Brunswick, Fredericton, NB, Canada, in 2004, and the Ph.D. degree in electrical and computer engineering from McGill University, Montreal, QC, Canada, in 2011.

He was a Professor with the Department of Geomatics Engineering, University of Calgary, Calgary, AB, Canada, from 2012 to 2024. He has been working as an Industrial Researcher with HERE Technologies (formerly NAVTEQ), Chicago, IL, USA, since 2008. He is currently a Distinguished Professor with Shenzhen University, Shenzhen, China. His primary research focuses on mobile LiDAR data processing for next-generation mapping and navigation.

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